SIEBEL CUSTOMER RELATIONSHIP MANAGEMENT (CRM) SUPPLEMENTAL GUIDE

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The materials presented in this technical guideline provide specific supplemental method guidance for Siebel Customer Relationship Management (CRM) implementations. The Siebel CRM Supplemental Guide contains the following components:

Siebel CRM Overview

This component provides an overview of Siebel CRM.

Siebel CRM Supplemental Task Guidelines

The Supplemental Task Guidelines should be used in conjunction with the standard OUM tasks in order to accomplish Siebel CRM implementations.

Siebel CRM Definitions and Terms

The Siebel CRM Definition and Terms should be used in conjunction with the standard OUM Definitions and Terms in order to understand terms referenced in this Supplemental Guide.

Siebel CRM Mapping

This component provides a mapping from OUM to the Siebel Results Roadmap (RR) Method.

PDF View

You may also prefer to access this information through the **Siebel CRM Supplemental Guide (PDF View)** located in the Supplemental Guides section of the Supplemental Guidance page.

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PURPOSE

The purpose of this document is to provide implementation teams with *supplemental guidance* for the successful planning and execution of Siebel CRM implementations using OUM.

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AUDIENCE

This document is intended for practitioners who will be implementing Oracle Siebel CRM solutions based on the OUM (Oracle Unified Method) approach.

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HOW TO USE THIS DOCUMENT

This document contains supplemental guidance for planning and executing Siebel CRM implementations and should be used in conjunction with and not instead of the OUM base method content.

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LIMITATIONS TO THIS DOCUMENT

This document does not describe OUM in detail – the reader must review the documentation and training materials referenced below to gain a basic understanding of each.

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RELATED DOCUMENTATION AND TRAINING

The following related documentation and training provide additional detailed guidance related to OUM implementations and are prerequisites to this document:

- OUM Method Pack – Desktop Installable or Hosted version
- OUM Training and Accreditation Program

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CONTRIBUTORS

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SIEBEL CRM OVERVIEW

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What is Oracle Siebel CRM?

Oracle's Siebel Customer Relationship Management (CRM) products cover the breadth of CRM functionality, from sales, marketing and e-commerce, to customer service, multi-channel interaction and analytics.

The following Oracle products are included within Siebel CRM:

- Sales
- Marketing and Loyalty
- Contact Center and Service
- Self-Service and E-Billing
- Contact Center Infrastructure
- Oracle CRM on Demand
- Partner Relationship Management
- Quote and Order Capture
- Price Management
- CRM Technology
- Social CRM
- CRM Gadgets

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ORACLE SIEBEL CRM AND OUM

As Siebel CRM is a packaged application, the following additional OUM Supplemental Guide should also be considered when conducting a Siebel Implementation project.

- Application Implementation Supplemental Guide

To assist practitioners in understanding how to employ OUM on a Siebel CRM project, the following content is provided to extend and complement the standard task and supplemental guidance indicated above. The Oracle Siebel CRM Supplemental Guide provides the following materials and tools to optimize the use of OUM.

- Task mapping between OUM and the previous methodology used for Siebel implementations (i.e., Siebel Results Roadmap) reflecting the set of recommended and

optional tasks required for a Siebel project)
- Siebel-specific content for tasks that require more detail for Siebel CRM implementation projects

Which Siebel Product Set Does this Guide Address?

This supplemental guide addresses Siebel CRM On Premises only. Oracle CRM On Demand, CRM Technology products, Social CRM and migration projects are out of scope for this version of the guide.

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SIEBEL CRM SUPPLEMENTAL TASK GUIDELINES

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This document contains OUM supplemental task guidance for Siebel CRM implementation projects.

- The first column of the following table provides a complete list of all tasks in this document for which "supplemental" guidelines are provided as well as a quick link to the "supplemental" task guideline in this document.
- The second column provides a link to the OUM Task Overview.

The task guidelines in this document should be used to supplement the guidelines that appear in the OUM Task Overview and the Application Implementation Supplemental Guide.

Siebel CRM - OUM Supplemental Task Guidelines (this document) collapse table | expand table	OUM Task Overview
Scope Management	
SM.010 Define and Confirm Scope	SM.010 Define and Confirm Scope
Business Requirements	
RD.001 Detail Business and System Objectives	RD.001 Detail Business and System Objectives
RD.005 Create System Context Diagram	RD.005 Create System Context Diagram
RD.011 Develop Future Process Model	RD.011 Develop Future Process Model
RD.015 Determine KPI Collection and Reporting Strategy	RD.015 Determine KPI Collection and Reporting Strategy
RD.042 Develop Glossary	RD.042 Develop Glossary
RD.065 Develop Domain Model (Business Entities)	RD.065 Develop Domain Model (Business Entities)
RD.140 Create Requirements Specification	RD.140 Create Requirements Specification
Requirements Analysis	
RA.023 Develop Use Case Model	RA.023 Develop Use Case Model
RA.024 Develop Use Case Details	RA.024 Develop Use Case Details
Mapping and Configuration	

MC.010 Define Business Data Structures	MC.010 Define Business Data Structures
MC.030 Map Business Requirements	MC.030 Map Business Requirements
MC.050 Define Application Setups	MC.050 Define Application Setups
MC.100 Define and Estimate Application Extensions	MC.100 Define and Estimate Application Extensions
Analysis	
AN.100 Prepare Analysis Specification	AN.100 Prepare Analysis Specification
Design	
DS.020 Define Application Extension Strategy	DS.020 Define Application Extension Strategy
DS.090 Design Data	DS.090 Design Data
DS.110 Design Business Rules	DS.110 Design Business Rules
DS.130 Design User Interface	DS.130 Design User Interface
DS.140 Prepare Design Specification	DS.140 Prepare Design Specification
DS.150 Develop Database Design	DS.150 Develop Database Design
DS.160 Review Design Model	DS.160 Review Design Model
Implementation	
IM.005 Develop Conceptual Prototype	IM.005 Develop Conceptual Prototype
IM.040 Implement Database	IM.040 Implement Database
IM.060 Perform Component Review	IM.060 Perform Component Review
IM.090 Create Installation Routines	IM.090 Create Installation Routines
Testing	
TE.020 Develop Unit Test Scripts	TE.020 Develop Unit Test Scripts
Performance Management	
PT.010 Conduct Performance Management Workshop	PT.010 Conduct Performance Management Workshop
PT.050 Design Performance Test Scripts and Programs	PT.050 Design Performance Test Scripts and Programs
Technical Architecture	
TA.010 Conduct Technical Architecture Workshop	TA.010 Conduct Technical Architecture Workshop
TA.030 Define Integration Requirements and Strategy	TA.030 Define Integration Requirements and Strategy
TA.150 Define Final Platform and Network Architecture	TA.150 Define Final Platform and Network Architecture
Data Acquisition and Conversion	
CV.020 Define Data Acquisition, Conversion and Data Quality Strategy	CV.020 Define Data Acquisition, Conversion and Data Quality Strategy
CV.025 Define Data Acquisition and Conversion Standards	CV.025 Define Data Acquisition and Conversion Standards
CV.040 Design Conversion Components (Initial Load)	CV.040 Design Conversion Components (Initial Load)

Transition	
TS.030 Develop Installation Plan	TS.030 Develop Installation Plan
TS.050 Prepare Production Environment	TS.050 Prepare Production Environment
TS.052 Implement Production Support Infrastructure	TS.052 Implement Production Support Infrastructure
TS.060 Go Production	TS.060 Go Production

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SCOPE MANAGEMENT (SM) PROCESS GUIDANCE

SM.010 Define and Confirm Scope

Templates and Tools

The following templates and tools specific for Siebel implementations are available to assist with this task:

Template File Name	Comments
SM-010_SCOPE_QUESTIONNAIRE.XLS	MS EXCEL

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BUSINESS REQUIREMENTS (RD) PROCESS GUIDANCE

RD.001 Detail Business and System Objectives

Approach

It is very important that this task is completed as it outlines the objectives for any Siebel implementation project. Upon completion of this task you should have a document listing all of the business and system objectives that the enterprise has identified to for the project so that they can be reviewed at various stages throughout the implementation. Ensure that each objective is numbered so that they can be easily be cross-referenced in other project documentation (i.e., the MoSCoW List). Also, the objectives should be placed into sensible groups to help locate the information at a later stage.

As a solution-driven approach, you generally review the documented business case and business benefits associated with the selected Siebel solution with the project stakeholders to confirm that they are the desired business benefits and determine how these objectives will be measured. The objectives should be related to the benefits offered by the solution. In the case of Siebel On Premises, the high-level benefits are increased customer satisfaction, gains in operational efficiency, and improved channel collaboration:

Gain	Accomplish
Strategic sales, service, call center, marketing, loyalty, partner management, customer order management and customer mastering capabilities	• Achieve maximum top and bottom-line growth. • Retain and grow customer relationships. • Increase loyalty and profitability. • Deliver personalized and consistent service experiences.

BEST PRACTICES FOR ESTABLISHING CRM BUSINESS OBJECTIVES

When defining your business objectives, consider the following best-practice guidelines:

Establish a clear vision for your Siebel implementation in relation to business drivers:

Your organization's vision for the implementation should address how using Siebel applications supports ongoing organizational goals. When establishing your vision, consider these questions:

- What were the critical business issues that prompted you to purchase the Siebel application?
- What does your organization hope to achieve?

The vision of your Siebel implementation is highly dependent on the business painpoints, challenges, hurdles, and issues you are trying to address. The following are a few examples of what may be driving your need to implement Siebel applications:

- Your customers are unhappy with the way they are treated when they interact with your organization through your call-center agent, field sales or service representatives, Web site, or partners.
- Your employees' activities are not customer-centric and therefore are ineffective in achieving your CRM objectives.
- Your partners feel that they don't have the information they need to truly act as an extension of your organization.
- Your call-center operations cost 200% more than the industry average.
- Your revenue per sales representative is 50% lower than the industry average.
- Your marketing, sales, and service functions don't share a common view of the customer.
- Your market share is being eroded by entry of new and more nimble competitors.
- New legalization has opened new markets to your business.

Example:

A medical instrument company rolled out a Siebel eBusiness solution to its sales representatives. The company defined a vision that was easy to understand and easy to articulate - to retain and acquire more customers. To achieve this vision, the company used the Siebel application to consolidate global call centers so that all used standardized and consistent processes for call handling, complaint investigations, and complaint reporting. Because of the clear vision established from the beginning, the many levels of the organization were aware of the "why" for the Siebel implementation and understood and supported the solution.

- Make sure that business objectives are specific, measurable, and achievable.
- Define metrics for measuring performance against baselines.
- Communicate your vision and business objectives across the organization.

Make sure that Business Objectives are measurable:

Review your vision for the Siebel implementation, then think about what you want your business to accomplish with your Siebel implementation. Do you want to increase revenue? Do you want to improve customer retention rates? The answers to these questions become the basis for your business objectives. Refer back to the selection process for your Siebel application, what were the ROI (return on investment) targets? For example, do you want to decrease costs, or increase productivity?

Make sure that your business objectives are specific and measurable such as "Reduce technical support response times by 20 percent" rather than "Improve technical support."

Make sure that you set realistic objectives. Your objectives need to be achievable so that your users will buy into achieving them. If you set the bar too high, no one will want to strive toward the objective.

Consider creating a hierarchy of business objectives, organized by business unit. This allows each business unit to establish how it will achieve its measurable business goals—which are linked to the overall business objectives. This way each business unit owns a part of the success of your implementation.

It is also important to make your objective time-bound. This helps you to define realistic objectives that lead to results. Clear, time-bound objectives make it easier to build

upon successes in each phase of your implementation. Achieving results bolsters confidence in your system and adds to a rapid receipt of ROI.

Examples:

Business Objectives	Non Business Objectives
• Reduce technical response time by 20% over a three-month period. • Increase forecasting accuracy by 15% during the first quarter.	• Implement CRM. • Replace a legacy system. • Improve technical support.

Define metrics for measuring against a baseline:

To get to your desired end-state, you need to determine and track specific metrics to gauge your progress towards meeting your business objectives. If you take a snap-shot view of how successful your implementation has been only at the end, you may risk being displeased with the results.

For example, you are deploying Siebel Sales and one of your business objectives is to increase the sales close rate by 15% in 12 months. In order to measure post rollout improvement, you need to establish a baseline of accurate close rates for your current (pre-implementation) sales process.

Careful definition of metrics and accurate baselining before rollout is critical for measuring the effectiveness of your implementation. There are other metrics that you need to measure as part your implementation, such as metrics that measure user adoption and metrics that measure leading indicators to give you early warning of potential problems.

The actual metrics most applicable to your business situation depends on your unique business objectives. However, the following table lists some revenue and cost-related metrics tracked by various types of world-class organizations:

Metrics	Marketing	Sales	Call Center and Field Service
Revenue	• Number of sales leads • Lead close rate	• Number of sales opportunities • Sales-focused activity time • Sales close rate • Partner performance	• Customer retention rate • Inquiry response time • Cross-sell and upsell rates
Costs	• Campaign cycle time • Campaign development cost • New product cycle time • Staff turnover • Staff training costs • System support costs	• Sales cycle time • Customer acquisition cost • Staff turnover • Staff training costs • System support costs	• Service cycle time • Fulfillment order errors • Problem-resolution rate • Staff turnover • Staff training costs • System support

Once you have determined what your metrics are, you need to baseline them. Many companies invest millions of dollars in state-of-the-art technology and rightfully expect high results. After rollout, however, these same companies, when probed to show exactly which parts of their operations have improved and by how much, struggle to deliver the numbers. This problem that can be avoided by gathering baseline data prior to the implementation, getting executives to agree on the current state, and then using the baseline to measure future performance and improvement. Consider the following examples, in which companies established a baseline and measured improvement against it:

- A large European telecommunications company increased revenue per customer by 20%.
- An international computer hardware company cut order processing time by 65%.

- A worldwide financial company improved lead conversions by 50%.
- A leading car manufacturer improved primary customer service metrics by 32%.

Communicate the Vision and Business Objectives:

Once you have established your vision and business objectives, you must communicate them throughout your company. Don't keep your vision and objectives secret or confined to executives and IT professionals; this creates a disconnect with the groups who are to support the implementation. Once you have established your vision and business objectives, share that vision broadly. When communicating your message consider the different reasons why those in your organization need to understand the vision and business objectives:

- The IT professionals need to understand what you want to accomplish, so that they can develop the best implementation approach and identify configurations that are necessary to bring about the desired results. Communicating the vision and business objectives helps your project team to focus the project scope and prioritize implementation decisions.
- Executives need to understand why the change is needed so that they can provide a business-centric, implementation strategic plan for business units.
- Managers use business objectives to understand the value of the new system early in the process and to communicate the benefits to users.

Lack of clarity in your vision statement can lead to user confusion. For example, one company issued a vision statement to "Improve the ability of our call centers to operate at a lower cost." The users interpreted this to mean "we are making significant job cuts" and went on strike. The reality was that the company had no intention of reducing the workforce. It wanted to increase call center efficiency to keep up with increasing call volumes. To achieve the desired results for your Siebel solution, you need to have everyone focused on and supportive of the objectives. Workshops can be used as forums where the implementation vision is communicated to frontline managers. Workshop participants get the same information at the same time, and feedback is encouraged. Workshops can also help the organization identify areas of resistance and can be used deliver a consistent and positive message. Newsletters, email announcements, and intranet articles are good methods to broadcast the vision and objectives to the entire user base, as illustrated in the following example.

Example:

The CEO of a software company succinctly defined the company's vision for its Siebel application and communicated its vision to employees by email. In the email, the CEO explained why the Siebel application was chosen and what the company expected to achieve from the implementation. The email assured employees that the CEO supported the initiative, was involved in the project, and would expect results from each employee during the rollout of the Siebel application. The CEO's first email was the launch of the communication strategy; it detailed the expected ways the organization would receive communication and provided a forum for feedback. This email was the first of many updates sent to the organization's employees to inform them about project progress and future plans. As important as communicating the message is encouraging two-way communication. After the initial email announcement was sent, the company set up an intranet site where employees were able to ask questions and post feedback, anonymously if they wished. The company committed to answer any questions within two days of posting.

Work Product

The Business and System Objectives work product describes and prioritizes the objectives as it is indicated in OUM Task Overview but should also indicate for each objective:

- Objective Type (driver or objective)
- Objective Executive Sponsor
- Organizations, related organizations and the sub-organizations under these (Examples: Sales Operations, Field Service Operations, Technical Support, Customer Service, etc…) involved in the objective.
- Channels (Examples: Live Telephone, Interactive Voice Response, Direct Sales, Direct Field Service, etc…)
- Products associated to each organization and channel

A number of example objectives are identified bellow:

- Improve operational costs.
- Increase first call resolution.
- Improve customer satisfaction.
- Decrease customer call back time.

- Increase customer service productivity.

Include in this work product also the related metrics, the related processes and sub-processes associated to each metric, numbering each metric and process as they will be cross referenced within further documentation. Examples of several metrics for one same objective:

- Related Metric: Decrease Service Resolution Time Related Processes. Related Processes: Manage Customer Retention, Review Service Request (Shared), Manage Inbound Service Call, etc…
- Related Metric: Reduce Average Speed of Answer. Related Process: Manage Inbound Service Call
- Related Metric: Increase First Call Resolution Rate. Related Processes: Initiate Service Request, Manage Inbound Service Call

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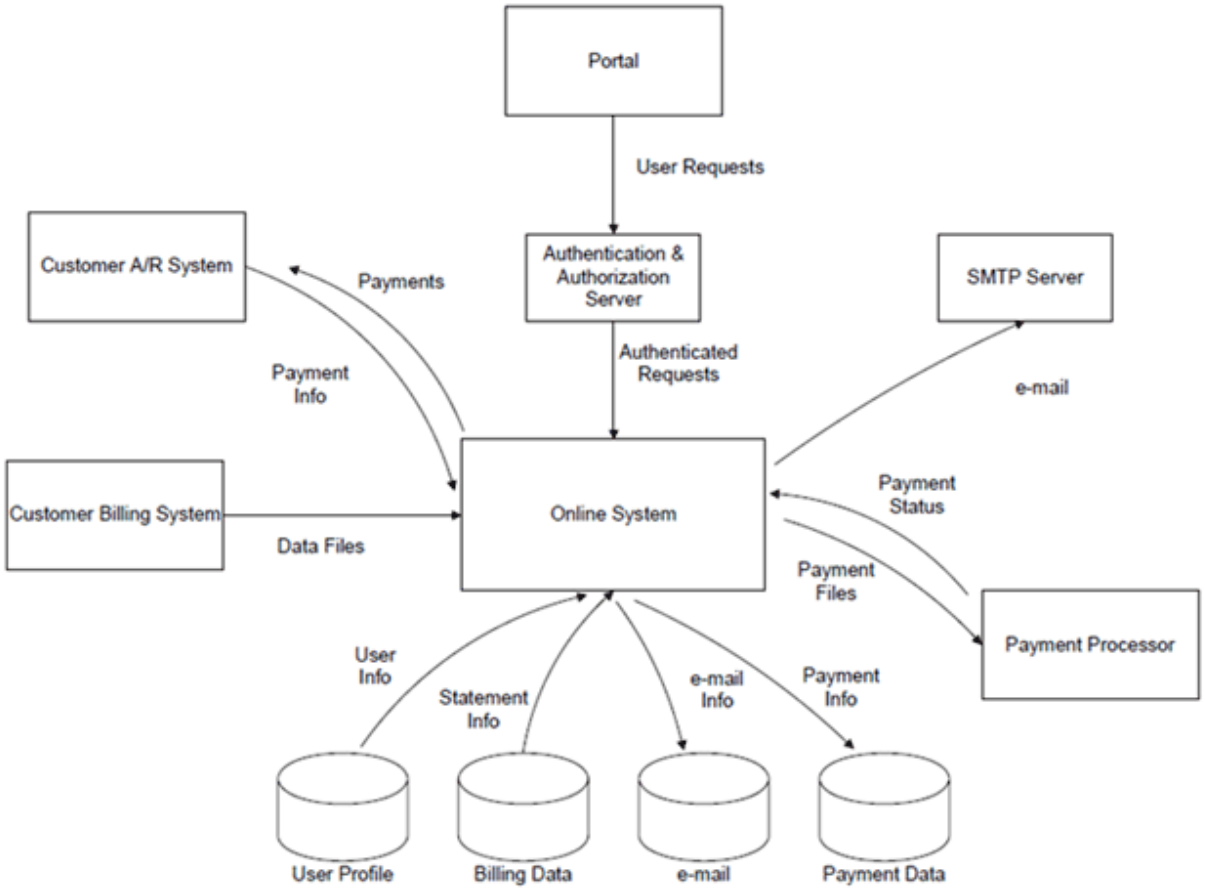
RD.005 Create System Context Diagram

The diagram of the main Siebel application functions, in the context of the entire system, is often contained in the consumer edition of the Application Guide for the corresponding Siebel application. This diagram can be used as a base to be adapted to the customer systems using MS VISIO.

The following image shows an example of a system context diagram for a Siebel implementation (Communications Billing Manager):



RD.011 Develop Future Process Model

Business Process Modeling Overview

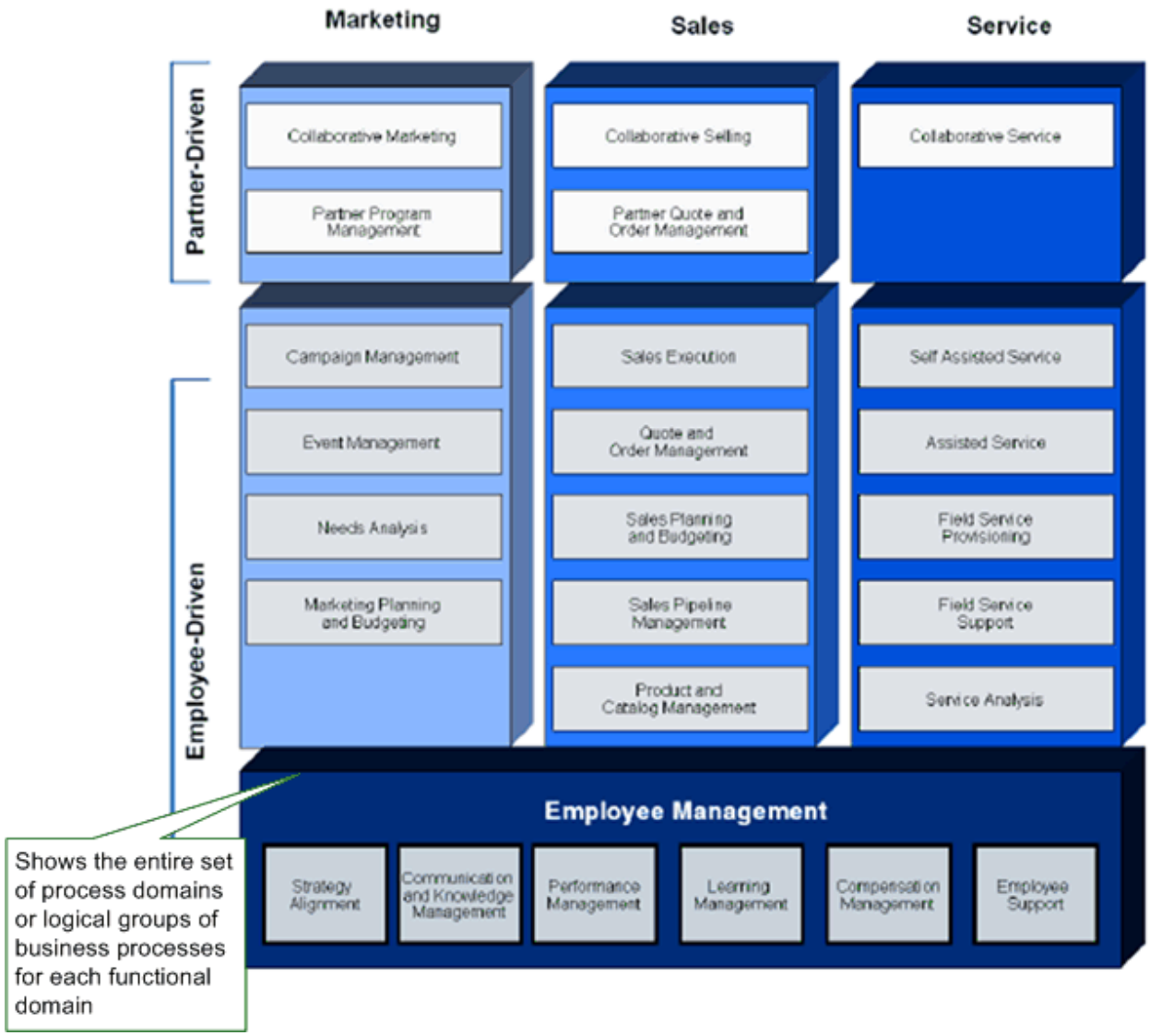
A business process model depicts the flow of work typically followed by users or systems to complete a sequence of tasks. Models can map business processes at multiple levels. Siebel implementations use four levels of detail:

1. The solution map shows the entire set of business processes for a single enterprise domain or multiple enterprise domains. An enterprise domain is a grouping of business processes from the perspective of an entire enterprise, such as the cross-industry set of business processes covering Customer Relationship Management (CRM), Employee Management, and Partner Management. The Solution Map also shows the functional domains within the enterprise domains. A functional domain applies to a typical business unit within a company, such as the Sales organization.
2. The process relationship diagram shows one functional domain, such as Sales, and the process domains with the groupings of related business processes within those process domains. A process domain, such as Sales Execution, groups business processes logically by activity. For example, the Sales Execution process domain includes business processes for Account Management, Lead Management, and Opportunity management.
3. The business process diagram shows a series of steps completed in the flow of a business process that accomplish a major business objective.
4. The subprocess diagram shows a greater level of detail for the high-level tasks represented in a business process diagram.

DIAGRAMS

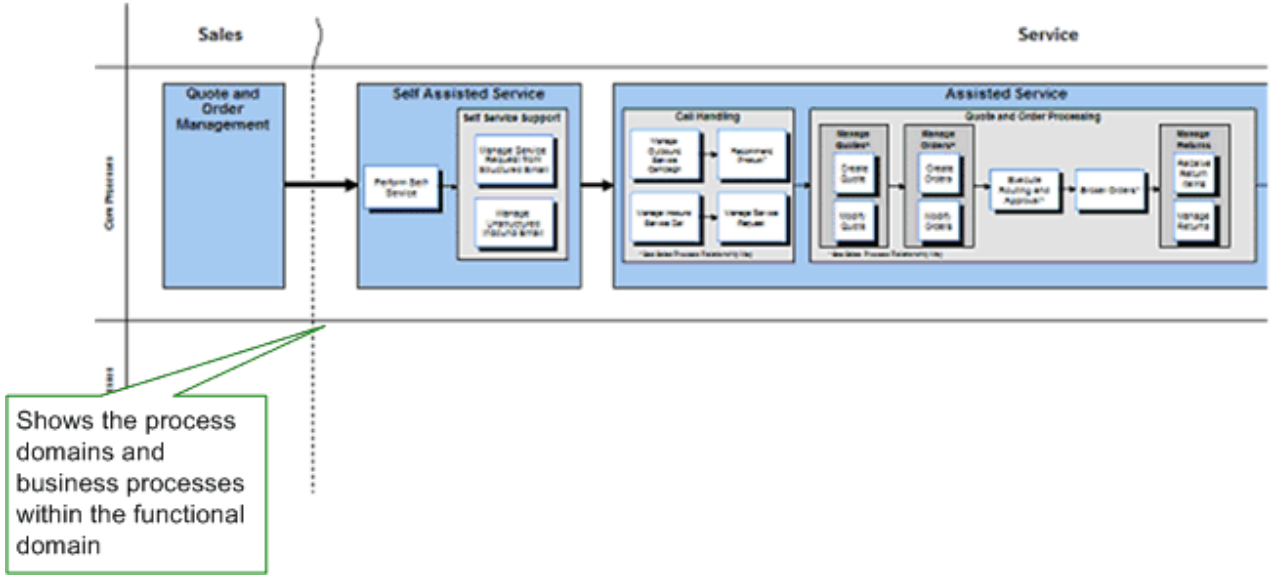
Solution Map

A solution map provides high-level business process models categorized by the business objectives of the organization. A solution map includes the process domains in which tasks are executed to meet these objectives. For example, a solution map for the enterprise domains of CRM, Employee Management, and Partner Management shows process domains including Marketing, Sales, and Service. In turn, the Marketing process domain shows functional domains that include Campaign Management, Event Management, and Needs Analysis. A business analyst or manager can review the solution map to make sure that the way a task, such as campaign management is performed supports the organization’s high-level business objective for marketing activities.



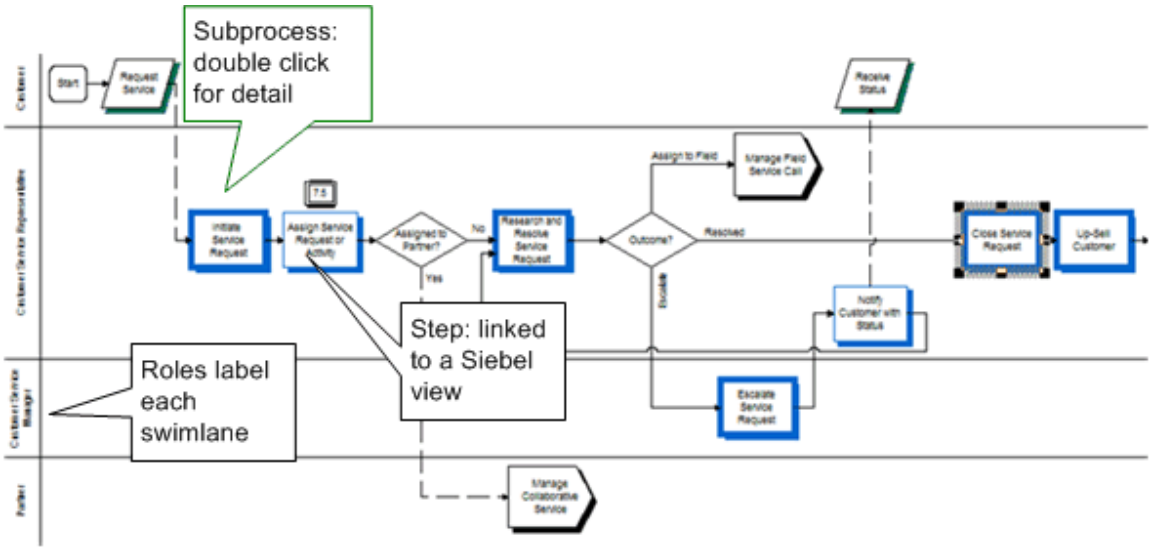
Process Relationship Diagram

The next level of process model, the process relationship diagram, represents more detailed business processes for one functional domain, e.g., Service and how it relates to other domains such as Sales. A process relationship diagram shows one functional domain, such as Sales, in detail on one page. Process relationship diagrams show the process domains within the functional domain. The process relationship diagrams also show how the business processes within a number of process domains relate to one another.



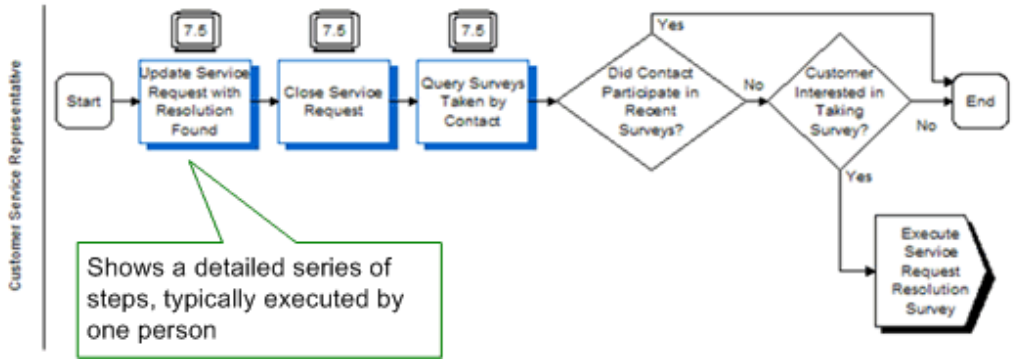
Business Process Diagram

The third level of detail, the business process diagram, shows the actual sequence of steps involved in a task referred to in a process relationship diagram. Within the Call Handling section of the process diagram there is a business process called, Manage Service Request. This process is show in the screen shot above. This level of diagram maps strategic processes, indicating interactions between users in your company and the end customer, as well as partners and external companies. This level contains steps and subprocesses, as well as clearly defined inputs and outputs.



Subprocess Diagram

You can apply even more detailed business process models, such as closing a service request. These processes are represented with subprocess diagrams. This diagram shows a series of steps, typically executed by one person that can be used in multiple business processes. Subprocess diagrams allow the detail of common work sequences to be summarized at the business process diagram level, so as not to detract from understanding of the overall function of the business process. Subprocesses detail the steps supported by the Siebel application, Universal Application Network (UAN), Business Integration Processes, and external applications.



BUSINESS PROCESS MODELING TERMINOLOGY

Oracle has codified both the functionality for common best practices for sales, marketing, and service processes and industry-specific best practices into our applications. In order to create or modify process solutions, business process modeling terminology must be reviewed. The table below summarizes some key modeling terminology.

Term	Definition
Functional Domain	A typical business unit within a company, such as sales, service or marketing
Process Domain	A logical grouping of processes, such as Sales Execution and Pricing Management, in a Functional Domain
Sub Domain	A subset of related processes in a Process Domain
Business Process	A sequence of steps and subprocesses performed to achieve a major business objective
Subprocess	A sequence of steps completed by an individual in one work session
Step	A single activity performed by an individual in executing a process
Business Integration Process	A built-in integration between a Siebel application and another enterprise-class application

MODLEING STANDARDS: ORGANIZING THE MODELS

As described in the OUM Overview for this task, if pre-defined process models/business flows exist for the functionality being implemented, they should be used as a starting point for preparation of this task's work product. Whenever possible, adapt existing standard models to the future business processes rather than creating business process models from scratch.

You can find Siebel-specific business process models, from an integration perspective, within the Oracle Business Process Architect Suite. This tool includes Oracle's conceptual business process models which were created by examining a variety of sources and combining them into a comprehensive set of best-practice business process definitions. Oracle E-Business Suite, PeopleSoft, JD Edwards, and Siebel business process definitions were combined into a superset of process information. The superset was then rationalized and compared to external authoritative and guiding sources for clarity and completeness. This superset is continually being updated and improved by knowledge gained with Oracle acquisitions and industry expertise.

There are four levels of models, L0-L3. L0-L2 models are high level conceptual models which can be used for Siebel applications implementations, whereas L3 models portray the process from a technical integration perspective not specified for Siebel.

You can find more information about the Delivered Reference Process Models (note 824633.1) available on My Oracle Support.

RD.015 Determine KPI Collection and Reporting Strategy

Measurement guidelines need to be identified to perform the analysis in at least the following areas:

- Key performance indicators / metrics
- Performance measurement dashboards
- Usage of systems
- Service delivery standards (e.g., up-time)
- Data quality standards

A formal KPI program can assist a company in measuring its progress and reinforce necessary change:

- Establishes a baseline from which to analyze impact on Sales, Marketing, and Services processes over time
- Ensures consistency of metric definition and reporting across business units
- Demonstrates quantifiable benefits early to gain buy-in and build momentum for the project
- Enables early and systematic identification of favorable as well as any unfavorable deviations from expected performance
- Facilitates problem identification and prompt corrective action, as appropriate

Typical KPIs include resource utilization (CPU, memory) of any server component, uptime, response time, transaction throughput, and reliability of your Siebel application while it is under a load of multiple users.

The Siebel system has been tested locally at all sites. Whenever you first log onto the system or first enter a tab, the performance is slower than when you subsequently work with that tab. This is because Siebel caches data to ensure that frequently used views remain in memory. The table below details typical timings to be expected doing different things in the system.

Task or Activity	Initial	Subsequent
Starting up Siebel (getting to login screen)	1-2 seconds	1-2 seconds
Logging in to Siebel (getting to the home page)	5-15 seconds	5-15 seconds
Clicking on any top tab (except Contact Summary)	2-5 seconds	1-2 seconds
Clicking on Contact Summary (this contains 8 views and connects to the CDB to retrieve data)	7-12 seconds	5-7 seconds
Running pre-defined queries (searching through and/or retrieving small data sets)	3-12 seconds	2-6 seconds
Running pre-defined queries (searching through and/or retrieving medium data sets)	20 seconds -1 minute	15-30 seconds
Running pre-defined queries (searching through and/or retrieving large data sets on complex search criteria)	Up to 10 minutes	Up to 10 minutes
In-line queries (searching through and/or retrieving small data sets)	2-5 seconds	2-5 seconds
In-line queries (searching through and/or retrieving medium data sets)	2-20 seconds	2-20 seconds
In-line queries (searching through and/or retrieving large data sets)	Up to 1 minute	Up to 1 minute
Send an email	Up to 1 minute	Up to 1 minute
Creating correspondence	Up to 1 minute	Up to 1 minute
Creating a report	Up to 1 minute	Up to 1 minute
Logging out of the system (Control+Shift+x)	2-5 seconds	N/A

It is possible that the performance experienced will change as more users come onto the system. This will be actively monitored so that remedial action can be taken. It is not possible to test every possible way of querying data in the system. As a result, it is possible that a particular query you create or run does not perform well.

Please report any instances of this, with as much detail as possible (e.g., the type of query you were running, the query parameters you were using), to your local help desk. Before you do so, please consider the quantity of data your query may searching through or returning. If either of these is relatively large, the query may take some time to execute.

Performance of email, correspondence, and report generation will depend upon the amount of data being generated. Note that large reports may time out and can be viewed in the Reports Server later.

RD.042 Develop Glossary

This task is highly recommended for all Siebel projects. It is essential that a project glossary be developed as it avoids misinterpretation of Siebel and business terms.

Refer to the Siebel CRM Definitions and Terms within OUM for a high-level Siebel glossary and the corresponding bookshelf for more specific terms.

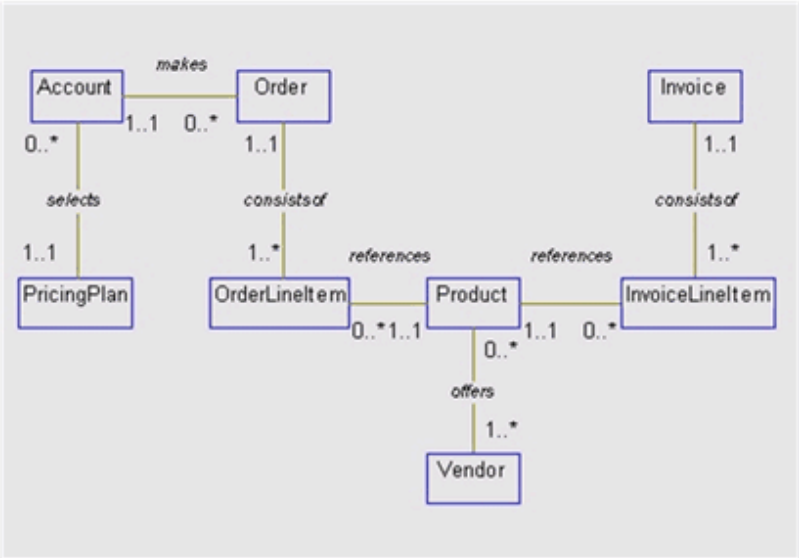
RD.065 Develop Domain Model (Business Entities)

This task has more than one iteration for Siebel projects:

Conceptual Object Model

The primary objective of this version is to establish a base understanding of the major business objects in the system. When creating this iteration of the model, the modeler is not concerned with object attributes, object operations, finding aggregations, or developing the inheritance hierarchy. However, it is useful to establish the association(s) between the major business objects, name the associations, and select their multiplicity.

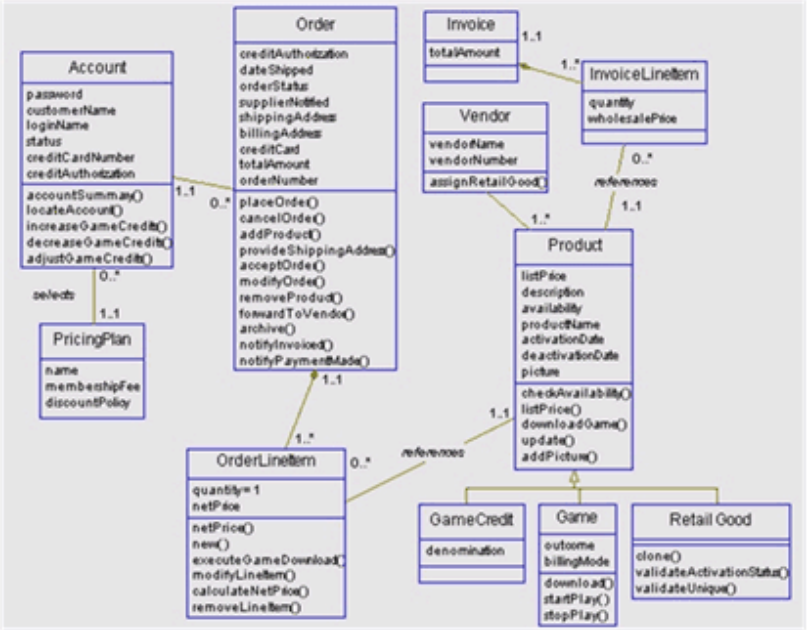
Example:



Logical Object Model

This version of the class diagram builds on the foundation established by the conceptual object model, and fills in many details, such as attributes and more refined associations such as aggregations. It also begins to consider the inheritance hierarchy. Depending on the style adopted by the modelers and the size of the application, it may take a number of iterations to refine the class diagram to the level of detail shown. As the design proceeds and other deliverables such as sequence, collaboration, and state diagrams are developed, the class diagram will be further refined.

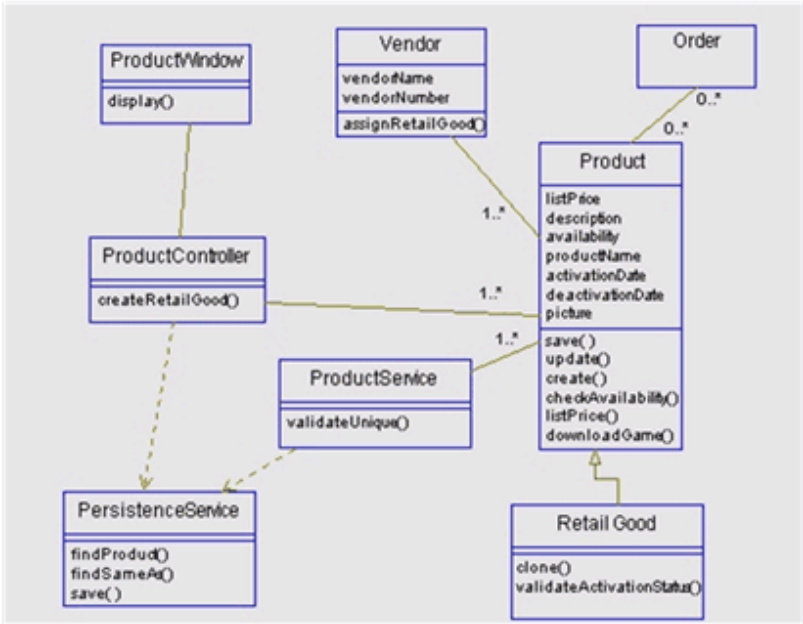
Example:



Physical Object Model

Updates to the behavioral model must be reflected in the structural model. This class diagram shows the new or updated classes, attributes, operations, and relationships implied by the updated behavioral model. Notice that the class diagram contains classes from other layers of the application such as the user interface and the database or storage layer, as well as a controller class. (A complete listing of classes would include all classes from each layer, for instance, all of the user interface classes, all classes encapsulating business logic not in the business classes, and so forth).

Example:



RD.140 Create Requirements Specification

Where the characteristics of a project support or require it, the RD.140 Requirements Specification may be used to consolidate a number of requirements-related work products. The Requirements Specification may contain both functional and supplemental requirements, forming a single requirements document for review and approval.

Functional Requirements

Functional requirements capture user or stakeholder requirements that describe tasks the users (actors) must be able to accomplish with the system in order to achieve stated Business Objectives; often referred to as a “wish List” of what users expect or desire of the system.

These requirements should be communicated and documented in user terms and language, and be void of technical terminology or jargon. Each requirement should describe a single desired result. As a result, conjunctions such as “and” “or”, “with”, and “also” should not be used in defining the requirement. Furthermore, each requirement should detail the following four items:

- User (Actor) – The person or thing (can be the system) that carries out the activity.
- Activity – Describes what task is being completed.
- Subject – Describes the subject of the activity.
- Qualification (optional) – Specifies any qualities, constraints, etc. that are associated with the activity being performed.

All user requirements should be associated/mapped to at least one role. This paradigm typically makes the sentence structure of the requirement statement “verb/noun”. Also, the requirements are generally listed as “ability” statements starting with “Ability to …” and should be concisely stated within 7-15 words. An example of a stakeholder requirement is, “ability to view customer list by account number”.

Additionally, functional requirements define the functionality the configuration team must build, or ensure exists, in the product that will enable users to accomplish their tasks, thereby satisfying the user requirements. These requirements provide the foundation upon which test cases are developed. They are generally stated as “shall” statements starting with “System shall…: For example, “System shall provide the ability to query for customer information by account number”.

Business Rule requirements can also be included as part of the functional requirements, as extension of these. They provide detail relative to the operating principles about the system, such as who can take what action and under what circumstances. Quite often, they will lead functional requirements that enforce them and thereby be mapped as such. An example of a business rule is: “only directors can approve projects greater than $500,000”.

Non-Functional Requirements

On the other hand, as part of the non functional requirements, it is very important in a Siebel implementation to gather the interface requirements used to capture the specific ones that are associated with integrating Siebel applications with other systems. These requirements should detail the following values:

- **Concurrency**
- **Conflict Resolution:** How system should behave if there is a problem? e.g., update of account from Siebel is sent off to an external system but account in this external system does not exist. What should system do?
- **Conversion Approach:** How will data be converted from old system to Siebel?
- **Frequency:** What is the frequency of the batch; how often will real-time interface be hit.
- **Interface Direction:** Siebel to other external system or external system to Siebel?
- **Interim Interface Approach:** How are you handling interface in the interim?
- **Volume:** Ties a gross number, e.g., 10,000 records over some period of time.
- **Integration Type:** Batch, Real-time Asynchronous, Real-time Synchronous, etc.

There are many other areas or types of non functional requirements involved in a Siebel implementation. Non functional requirements are generally stated as “shall” statements starting with “System shall…”

Administrative Requirements: Requirements captured to set up positions and responsibilities within the Siebel Application.

- **Availability/Scalability:** Requirements gathered to facilitate continuous operation of servers, applications, databases, and network services in a distributed environment. These document the architecture requirements for the system based on transaction volumes and user types. Availability and scalability requirements should contribute to providing network users with continuous access to resources that are required for them to perform their jobs without disruption.
- **Data Conversion:** Requirements that describe existing functions that need to be converted in order to provide the complete business solution in the new system. These requirements describe how the system accomplishes its functional requirements.
- **Data Migration:** Describe the way the existing data will be converted or migrated to the new system.
- **Data Cleansing:** Existing data that must be transformed and propagated to enable migration to the new system. The intended meaning of the data and its quality must be known to complete this analysis. The task should result in the discovery of data anomalies, errors, suspect data, and actual business rules that have been applied to the data.
- **Data Dependency:** Identifies objects, object attributes, and required relationships within the data that need to be managed.
- **Deployment:** Identifies requirements that define when and to whom the system is deployed.
- **Localization:** Identifies language, currency, and time zone requirements.
- **Performance:** A Performance requirement imposes conditions on functional requirements. It may specify performance parameters for speed, efficiency, availability, accuracy, throughput, response time, recovery time, or resource usage. These requirements describe how the system accomplishes its functional requirements in the areas of speed and memory requirements. Usually accompanied by a Sizing Review.
- **Pricing:** Related to requirements specific to ISS implementations. Creation of detailed pricing algorithms to support different purchasing behavior, etc.
- **Products:** Related to requirements specific to ISS implementations. Establishment of product catalogs, etc.
- **Security:** Identifies user requirements in the areas of system access (logon), data access, and business function access.
- **Technical:** Defines requirements in the areas of database, repository, network, server, client, and system topology to support the underlying software implementation. Focus is placed on the identification of key technical components relative to the implementation of Siebel in a complex organization.

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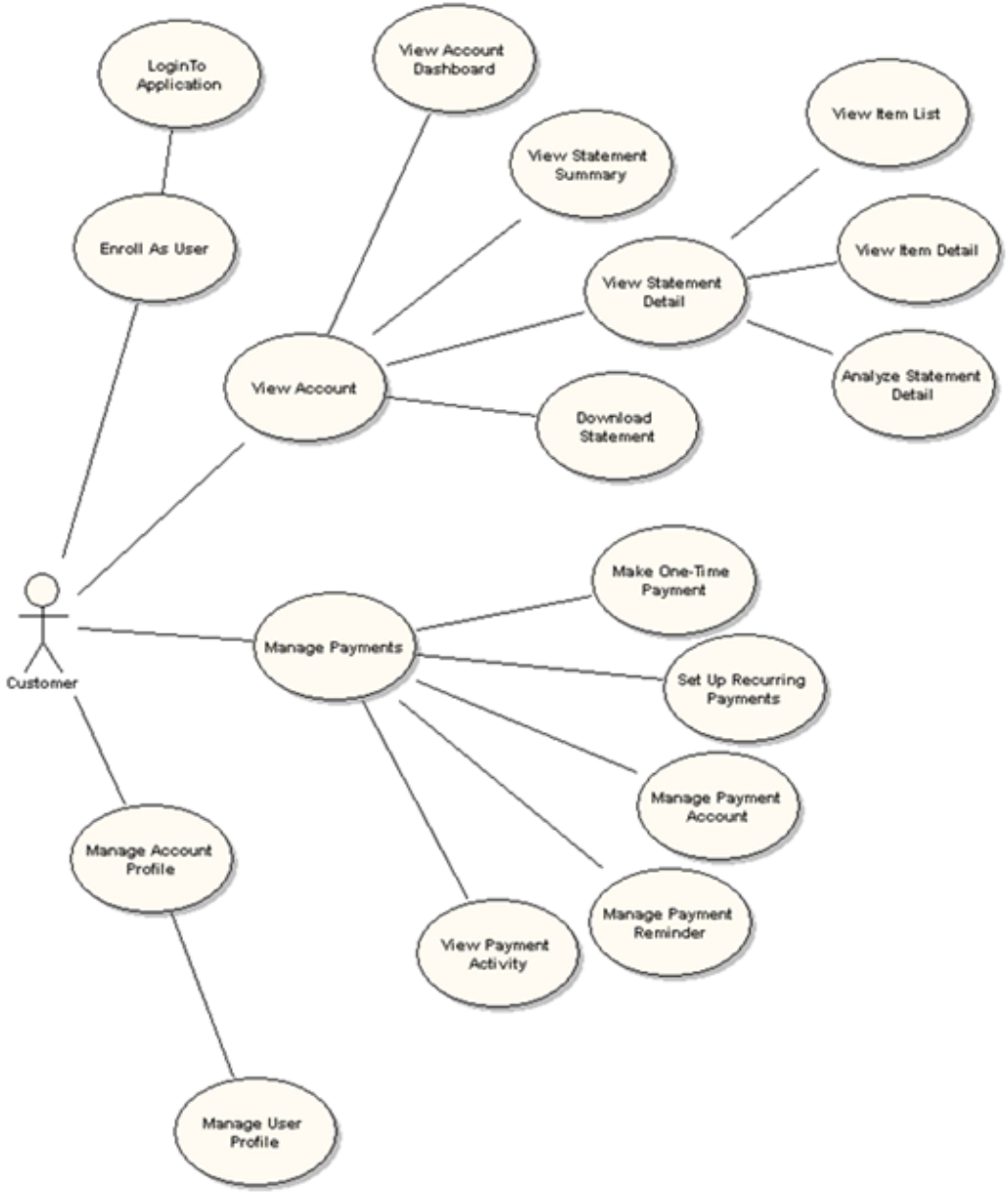
REQUIREMENTS ANALYSIS (RA) PROCESS GUIDANCE

RA.023 Develop Use Case Model

As described in the baseline Overview for this OUM task,“At a minimum, use case models should be developed for those requirements that can only be satisfied through custom-built components, which extend the functionality of the COTS system and/or integrate the COTS system with other COTS or legacy systems. Development of a Use Case Model for all other business requirements is considered best practice, as they contribute to a better understanding of requirements and can be leveraged to prepare a number of other required work products downstream, such as test scenarios and business procedures documentation.”

In Siebel implementations, it may also be advisable to create a use case model for requirements satisfied by Out-of-the-Box (OOTB), standard functionality. The consumer edition of the Application Guide for the corresponding Siebel application contains the Catalog of Applications. This is a diagram showing how the use cases relate back to the overall application logic and navigation. This diagram can be used in the case where use cases are developed for all requirements. They can also be used as a base to be adapted to the specific requirements of the project.

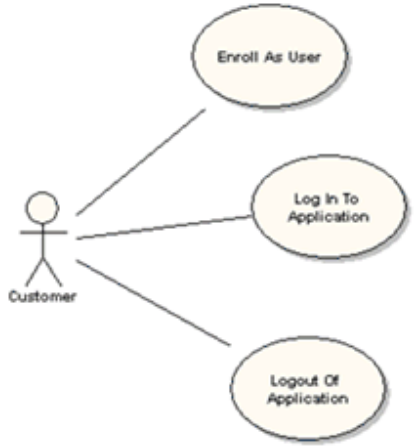
Example:

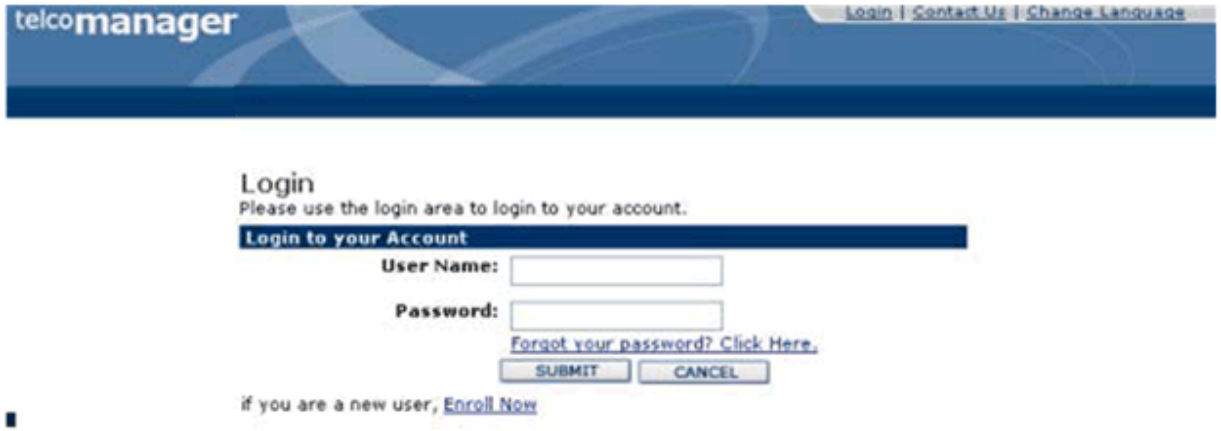


A list of the web flows that specify the functionality and site navigation of the application and the use cases covering all the OOTB requirements is also included in the consumer edition of the Application Guide including screen shots of the navigation.

Example - Authentication Use Case:

The authentication use case covers the business requirements for allowing customers to access the system. This use case allows consumers to login to the application and access the system as an authenticated user.





Detail	Description
Name:	Login to application.
Brief Description:	Consumer user logs into the application.
Primary Actor:	Consumer user
Main Path:	1. This use case begins when a user navigates to the login page. 2. System displays Login page displaying a form for entering user name and password. 3. User enters login information. 4. User submits data. 5. System validates data.
Alternate Paths:	3a) User has forgotten their password and selects the Forgot Password operation: 1. System invokes Forgot Password use case. 3b) User has not enrolled and selects the Enroll Now operation: 2. System invokes Enroll User use case. 5a) System determines that customer credentials are invalid: 1. System displays a message “Login incorrect: Please try again”
Standard Features:	1. Authentication of User Name and Password 2. Forgot Password link
Configuration Points:	1. Messaging instructions around the login form. 2. Communications Billing Manager ships with a default enrollment and authentication model for user management. This framework for user management may be configured via a plug-in to use available systems for authentication and other user information.

Notes:	1. Default implementation stores the enrollment information within the Communications Billing Manager application.

Using Use Cases with Packaged Applications

There are several things to keep in mind when using use cases with a packaged application like Siebel. Under most circumstances, use case methodologies suggest that the business requirements be completely separated from the GUI design. Furthermore, the UML-defined efforts surrounding the completion of Use Cases and particularly the steps for design and development almost always assume the system is being built from a blank slate. When managing use cases for packaged applications, there are some very important considerations. Most importantly, we can assume that packaged applications have already been built, and they are being customized or configured by an implementation team:

1. Terminology for things (GUI elements, objects, data) already exists.
2. Business processes may be imposed or defined by the application and thereby tie requirements and OOTB functionality to a strict process.
3. The technology is implemented within a certain framework.

Those facilitating requirements will constantly iterate through understanding how the enterprise's current terminology, process, or application maps to Siebel.

Terminology in Packaged Applications

The application already has names for things – in discussing the application with business users, it will be important to explain the key terminology used within the application and begin to align those pre-defined items with the terms the business uses. For Siebel applications, this could be terminology from a business process, user action on an item, functional area, GUI element, etc.

- Business Process Examples – Pipeline Management, Order Management, Forecasting, etc.
- User Actions Examples – Merge, delete, query, drilldown, scroll, etc.
- Functional Area Examples – Agreements, Accounts, Service Requests, Activities, etc.
- GUI Element Examples – Screen Tabs, View Tabs, Queries Drop Down, Show More, etc.

When capturing use cases, those responsible for capturing use cases should be careful to introduce terms appropriately to allow the user to define their goals in their own terms while still matching the user's terms to Siebel terms. Users that are familiar with Siebel will be more likely to describe their goals in Siebel terminology. As well, the use case and requirements document templates should take terminology into account.

RA.024 Develop Use Case Details

Here are additional things that can be derived or included in this work product for each use case:

- **Actor:** is a user type and the name represents an action (verb) on an entity or related entity (noun). Nouns include key entities within Siebel and verbs include Navigate, Query, Create, Relate/Associate, Disassociate, Delete, and Merge.
- **Scope:** This list can be used to determine what will be designed, developed, and deployed for a set of project phases, projects, or even project teams. It can drive accountability and awareness amongst development areas that overlap within a team or across teams.
- **Functional Organization:** The Actors mapping to Task-Level Goals can define whether sales, service, or marketing organizations use the application. The names of the Task-Level Goals include nouns that represent entities – these can be grouped together by Accounts or Service Requests. The verbs are key actions (Query, Create, Edit, Delete, etc.) that are found on most applet menus and represent things that users do to records. Additional actions are outlined below.
- **Categorization of Use Cases:** Grouping, categorization, and meta-classification of use cases is important and includes the ability to place value on a group of use cases, associate phases and package their delivery (things that must be delivered together), assign level of effort, or even associate them to types of development like workflow or scripting.
- **Relationships to other Use Cases:** Siebel provides a very clear and consistent framework for relating entities (1:1, M:1, 1:M, and M:M). These relationships are

established via Form Views (Parent Form Applet, Child List Applet), Pick Applets, and MVG Applets. Furthermore, there can be multiple relationships, bidirectional, and different types of relationships between two entities. Examples:

- 1:1 or M:1 – Entities can be Created and Related OR just Related:
  - 1:1 – Create (and Relate) Account's Profile: An Account-Profile view has two form applets. Create and Relate (Siebel can allow the creation of the profile form and relate it to the Account in the background by creating a foreign key)
  - M:1 – Relate Profile to Parent Account or Create Profile: An administrative All Profiles view has a pick applet for a user to pick an account for a single value field, relating the profile to its parent account.
- 1:M – Entities can be Created and Related OR just Related:
  - 1:M – Create (and Relate) Account's Opportunities: Account-Opportunities view has a form and list applet where new opportunities can be created – perhaps every child opportunity has a required field that is a foreign key relationship to its parent.
  - 1:M – Create Account's Affiliated Account: Account-Accounts View has a form and list applet where new account relationships can be created, but the child list applet has a pick applet to create relationships and does not allow the creation of new accounts.
- M:M – Entities can be Related OR Created and Related:
  - M:M – Relate Account's Address: An Account-Address view could have an MVG Applet where addresses can be related, and the new button (if available) spawns the association applet which allows for the creation of a new address record that also gets related to the account upon creation.

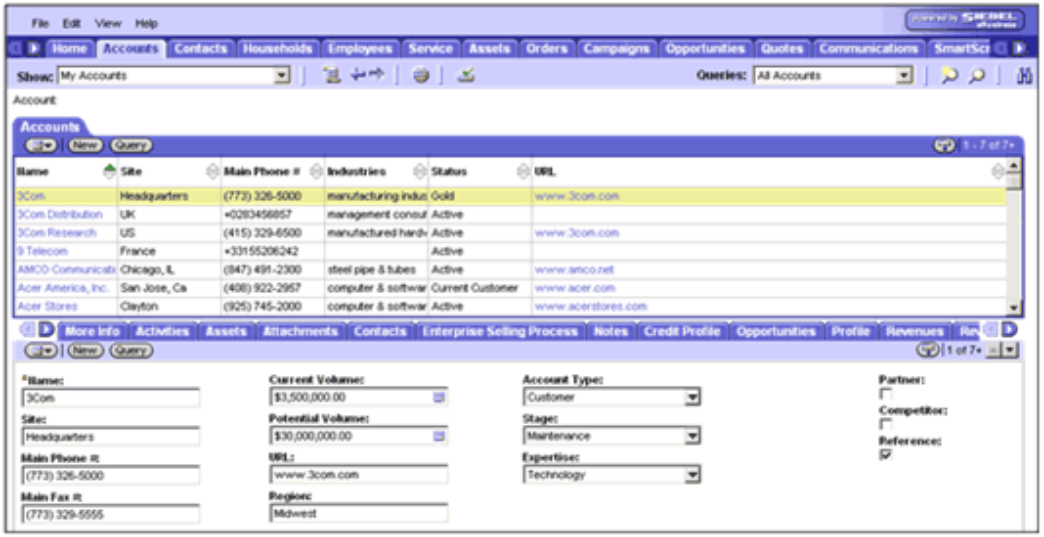
- **Development Effort per Use Case and Relationship to Siebel Objects:** This list of use cases can imply one or many views and/or applets. Considering the Siebel architecture, one can infer the business, data layer, and even integration components. For example, naming entities such as "Accounts" or "Account's Opportunity" also implies the use of certain views, objects, and tables in Siebel. These objects can also be called out in an additional column. Knowing this, designers and developers can easily provide high, medium, and low effort estimates to use cases. Similarly, it is easy to identify objects that are being used, conflicting, or overlapping their association with multiple use cases. Care should be taken because situations often arise where an applet's visible functionality (within a use case) is different under a different context of use (another use case) and the Siebel GUI reuses many objects. Therefore, the design should appropriately call out whether the object might need to be copied (for a different search spec) or where another aspect of the architecture is used (personalization).

Example of Use Case - Creating an account record

This example shows how a use case might be defined in the Use Case Specification work product template.

Context of Use

As a central object for managing customer relationships, an account record must be created for each prospective or customer account before other supporting information can be entered about the account. When the account record is created, the agent can begin to add and track important details such as contacts, opportunities, and service requests.



Main Success Scenario

What the Actor Does	What the System Does
1. The user navigates to the **Accounts** screen.	
2. The user clicks the Show drop-down arrow and selects **My Accounts**.	
3. The user clicks New in the **Accounts** list.	4. The system displays **New Account** mode.
5. The user completes the following required fields: Note: Click to expand the form and view all available account fields.	

Field	Description
Name	Type the business name of the customer account using title case, for example, Fred Smith Pty. Ltd.
Site	If an account has multiple locations, the Site field can be used to determine a more specific location.
Main Phone #	Type the account's central phone number, including the area code, with no spaces. The phone number is automatically formatted in (AAA) NNN NNNN format.
Main Fax #	Type the account's main fax number, including the area code, with no spaces. The phone number is automatically formatted in (AAA) NNN NNNN format.
Current Volume	Click and use , and to select **Currency Code**, **Exchange Date** and **Amount** for the current volume of business with this account.
Potential Volume	Click and use , and to select **Currency Code**, **Exchange Date** and **Amount** for the potential volume of business with this account.

URL	Type the full web address of any web site maintained by the account, for example, *www.geoffreysteel.com*.
Region	Type the name of the region, for example, *Midwest*.
Account Type	Select the account type that best describes the relationship between the account and company, for example, *Prospect, Consultant*.
Stage	Click to select the stage of the account, for example, *Rollout*.
Expertise	Click to select the expertise of the account, for example, *Technology*.
Partner	Click to indicate this account is a partner.
Competitor	Click to indicate this account is a competitor.
Reference	Click to indicate this account as a reference.

MAPPING AND CONFIGURATION (MC) PROCESS GUIDANCE

MC.010 Define Business Data Structures

As this task is oriented to Financials applications most of the content described in the OUM Task Overview is not applicable to Siebel implementations, so, this activity is not generally executed in a Siebel CRM implementation.

MC.030 Map Business Requirements

Approach

PREPARE FOR DETAILED GAP ANALYSIS

Although substantial information regarding the proposed solution should be available from earlier documentation, it will still be necessary to understand in significant detail the purchased functionality set of the customer's configuration so a thorough and informative gap analysis can be completed. As a result, if the purchased modules have not been loaded on a machine through which the functionality can be verified, this activity should be completed now.

The business analyst must understand the purchased Siebel functionality before conducting discussions with the enterprise's SMEs and end users regarding solution options. This is necessary to help understand which features and functionality will be implemented, to identify potential gaps, and to suggest alternate solutions.

Once the business analysts (and other members of the project team) understand the purchased functionality, they will be in a better position users to modify business processes, rather than modifying the Siebel application. Minimizing modifications will help to hold down configuration costs and increase the ROI.

COMPARE REQUIREMENTS AND PROCESS FLOWS TO APPLICATION CAPABILITY

The purpose of the mapping work is to map every requirement to the purchased Siebel functionality and identify potential gaps. The following steps should be followed:

- List the “TO BE” business processes and reference numbers. The enterprise should encouraged to optimize their current business processes to take advantage of the Siebel technology. If the enterprise chooses to re-engineer their current business processes, the recommended approach would be to map the existing business processes using the Siebel standard business models to create new “TO BE” business processes that will be most closely aligned with the standard Siebel functionality.
- List the business unit leveraging the business processes
- Reference the appropriate high-level business processes
- List any sub-processes that may exist to further define the “TO BE” business process.
- Provide a description of the detailed business requirement.
- Create a new business process flow (if applicable). If a robust process modeling and requirements management environment are available, it may be preferable to build upon the high-level process flows created during the implementation planning effort. Ideally, the standard process models should be used as the baseline materials that will be used to develop the software solution. Using these tailored models will better ensure that purchased functionality is optimized and any deltas from standard functionality are effectively tracked.
- Capture/list the base Siebel function corresponding to the business process. For example, CTI would solve the Call routing/VRU business process.
- Identify whether the Customer business processes that have been developed map to the purchased base Siebel functionality. If any deltas exist, flag these items as usage, functional, or capability gaps. Usage Gaps require light to moderate configuration of standard Siebel functionality, Functional Gaps involve adding or significantly enhancing standard Siebel functionality, and Capability Gaps cannot be supported within Siebel. More information into gap types will be discussed in the “Determine Rating Scale” section later in the document.
- Provide a description of any identified gaps.

Templates and Tools

The following templates and tools specific for Siebel implementations are available to assist with this task:

Template File Name	Comments
MC-030_MAPPED_BUSINESS_REQUIREMENTS_WORKBOOK.XLS	MS EXCEL

MC.050 Define Application Setups

Once your Siebel products are successfully installed, you need to perform a number of tasks to set up and administer the application. Initial setup tasks are those which must be completed in order for your Siebel applications to work correctly, but are then revisited infrequently, for instance, as your company grows. Ongoing tasks are done on a more frequent or ongoing basis. Both types of tasks may be accomplished by following the *Application Administration Guide*.

Initial Setup Tasks

Determine access control strategy and define business environment structure:

Siebel applications can be set up to support many strategies for your company to control access to views and data. These strategies include methods such as defining your business environment structure (organizations, internal and external divisions, and so on), defining employee positions, and creating access groups so that specific groups of people have access to specific views and data. These decisions should be made early in the deployment process, so that the strategy can be implemented during the initial setup. For more information about controlling access to views and data, and the procedures for implementing access control, see the *Security Guide for Siebel Business Applications*.

Enter employee records into system (database and application) and determine employee access to views and data:

Enter employee records after you have defined your business environment structure. You need to assign at least one responsibility to each employee, and you may also assign organizations, positions, or other access control parameters. For more information about entering or deactivating employee records, see the *Security Guide for Siebel Business Applications*.

Ongoing Application Administration Tasks

These tasks set up defaults that are used throughout your Siebel application as system preferences, predefined queries, currencies, currency conversion, expense types, payment terms, periods, telephone formats, pager companies, date formats, ZIP Codes, global time zone support, email, fax, mail accounts, industries, languages, case sensitivity, web browser capabilities, additional web browsers, quick fill templates, default view links for screen home pages, etc.

For some implementations, the default settings provided through the seed data meet a company's needs without customization. In other cases, administrators find it useful to make adjustments to these defaults.

Templates and Tools

The following templates and tools specific for Siebel implementations are available to assist with this task:

Template File Name	Comments
MC-050_APPLICATION_SETUP_DOCUMENT_SIEBEL.DOCX	MS WORD
MC-050_OUM_ADMINISTRATIVE_DATA_WORKBOOK_SIEBEL.XLS	MS EXCEL

MC.100 Define and Estimate Application Extensions

Approach

DETERMINE RATING SCALE

Once gaps have been determined to exist, the complexity and level of each gap needs to be assessed. This portion of Gap Analysis work is completed in user review sessions.

There are different types of gaps identified in the spreadsheet tool below that will directly affect the score of the gap complexity.

- **Usage Gap** – A gap with light to moderate configuration surrounds the main issue.
- **Functional Gap** – A gap that needs significant configuration to the base Siebel functionality.
- **Capability Gap** – there is not a Siebel solution

Once the type of gap has been identified, the gap will need to be quantified with a score based on a rating scale ranging between 1 and 5. A level 1 indication describes an "out-of-the-box" feature or very little customization while a level 5 indicates a very complex customization.

To improve the accuracy of the work effort estimates, separate gap ratings are provided for each of the five major categories of customization. Thus, a single requirement may have more than one gap rating.

Templates and Tools

The following templates and tools specific for Siebel implementations are available to assist with this task:

Template File Name	Comments

MC-100_APPLICATION_EXTENSION_DEFINITION_AND_ESTIMATES_WORKBOOK.XLS	MS EXCEL

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ANALYSIS (AN) PROCESS GUIDANCE

AN.100 Prepare Analysis Specification

Where the characteristics of a project support it, the AN.100 Prepare Analysis Specification may be used to consolidate a number of Analysis work products. The Analysis Specification can be used to create a single document that describes the overall functionality that must be provided by an extension or customization.

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DESIGN (DS) PROCESS GUIDANCE

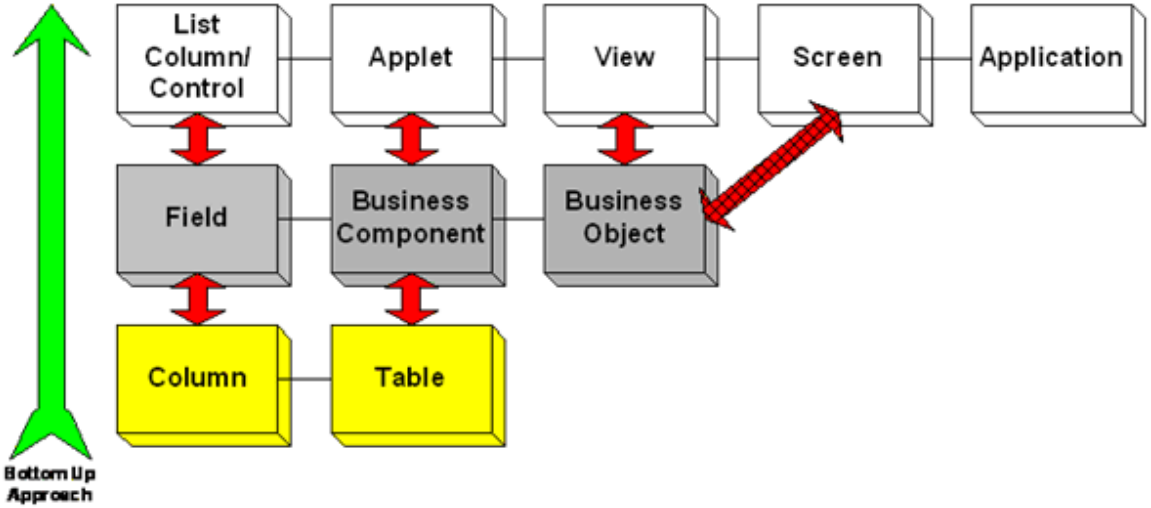
DS.020 Define Application Extension Strategy

Check the *Expert Services Development Guidelines and Configuration Concepts* document to ensure that any configurations you perform:

- are accurate
- provide an intuitive user interface
- are easy to maintain and upgrade
- maintain optimum performance of the Siebel eBusiness application

Some of these guidelines and concepts are related to:

- Repository Management, naming repositories: A consistent naming convention ensures successful configuration and testing and minimizes the work required when migrating new repositories or performing upgrades.
- Creating new objects:
  - Modifying existing object versus creating a new one: modifying existing objects is typically a preferable method because it requires less configuration work and causes less problems with future upgrades. Reducing the number of repository objects that will be evaluated/upgraded reduces the effort required for upgrading the application.
  - Use a bottom up approach: For Siebel development it is recommended that you create objects in a particular order. This will ensure that the correct values for all required properties are available as options. For example, before creating a business object, the business components and related links should be created. You should create all your data manipulation objects before you create your presentation objects.



- Disabling unused screens/views: Disassociate all views on the redundant screen from those responsibilities used by your organization. This approach reduces the amount of configuration that you need to maintain and upgrade.

Naming Conventions for Siebel Configurations

Use the following naming conventions for Siebel configurations:

Object	Convention
Any	For any new object that you create (top-level or child), prefix the name with the company name (or an abbreviation of it). For example, if you are creating a new Hobby business component for ABC Corporation, name it ABC Hobby. Similarly, if you are creating a new 'Description' field in the standard Account business component, name it ABC Description. Another approach to consider would be to put the three-letter acronym at the end of the name, instead of at the beginning. This convention will allow you to search for custom objects by acronym but still see related objects sorted together, such as Approval Authority and Approval Authority Code - XYZ The only exception to this standard is that if you are adding a new child object to a parent object that is already prefixed, there is no need to prefix it again at the child level. For example, if you are adding a Description field to the ABC Hobby business component, name it Description instead of ABC Description.
General	The majority of object names (except applets) should be singular, present tense nouns. • Avoid special characters like slashes, parentheses, periods, etc. in object names (except for slashes in link names). The "#" (number sign) character is a valid character. • Avoid prepositions in object names. For example, use # Weeks or Weeks instead of Number of Weeks. • Avoid the word Amount in the names of currency fields. For example, use Revenue instead of Revenue Amount.
Links	Adhere to the convention of <parentBusComp>/<childBusComp>. By default, the name for new links is set to this format, but it is possible to override the default name.
Join	When a new join is added to an existing standard business component, always give it an alias and ensure that the alias is prefixed with the company name or acronym.
Foreign Key Fields	Foreign key fields should have a name like <Entity> Id. If the foreign key is serving as a primary pointer, it should have a name like

	Primary <Entity> ID. For a primary pointer to an entity that is already prefixed, prefix the foreign key field with the company name or acronym (as usual), but do not include the prefix in the <Entity> name. For example, a primary pointer to the ABC Subsegment business component would be named ABC Primary Subsegment ID, rather than Primary ABC Subsegment Id or ABC Primary ABC Subsegment ID.
New LOV_TYPE in List of Values	Prefix with Company name or Acronym
Form Applets & Profile Applets	Titles should be singular.
List Applets & MVG Applets	Titles should be plural.
Associate Applets	Titles should be of the format Add <entities>, where <entities> is plural.
Pick Applets	Titles should be of the format Pick <entity>, where <entity> is singular.

- When naming currency code and exchange date fields, simply call them Currency Code and Exchange Date if they are the only such fields in the business component. If there are multiple instances of similar fields, prefix each with the name of the corresponding Amount column, for example, Revenue Currency Code and Budget Currency Code. The reason for this is that they are referenced by other fields when specifying the Properties Currency Code Field and Exchange Date field. By defining the fields in this way, it makes the reference easier to understand.
- The field URL needs to be called 'URL' and the class of the Business Component needs to be set to CSSBCBase, in order for the hyperlinking functionality to work correctly.
- Never include a question mark at the end of a field name or UI label.
- Use meaningful, descriptive object names (for example, Account Detail Applet With Agreement, rather than Account Detail Applet 2).
- Be careful about spelling, spacing, and capitalization when naming objects. Typically, logical names of objects in the repository use complete words, mixed casing, and spaces between words. On the other hand, physical database objects use abbreviations, uppercase, and underscores. For example, the Service Request business component is based on the S_SRV_REQ database table. Also, note the unusual capitalization of the word PickList as used throughout the standard repository.
- It is very cumbersome to change the name of an object after it has been referenced throughout the repository. If you ever do indeed need to change the name of an object that may have many references throughout the repository, use the Find in Repository feature (from the Tools menu in Siebel Tools) in order to find all those references.
- Read-only applets should always have the string Read Only immediately before the word Applet in their name. For example, ABC Account List Read Only Applet rather than ABC Account List Applet – Read Only.
- Read Only views should always have the string Read Only immediately before the word View in their name. For example, ABC Account Common Profile Read Only View rather than ABC Account Common Profile View – Read Only.
- Applets that duplicate other applets but do not have drilldowns should have the same name as the original applet but with the words Without Navigation appearing immediately before the word Applet (or Read Only Applet). For example ABC Selective Account List Without Navigation Applet.
- Applets that are used specifically for Administration purposes (which are almost always list applets) should be named <entity> Administration Applet. For example, Master Forecast Administration Applet.
- Business components that are used to represent child entities should not have their parent entity in their name. For example, ABC Subsegment rather than ABC Account Subsegment. Similarly, the applets that are based on these child business components should only reflect the name of the business component itself. For example ABC Subsegment List Applet rather than ABC Account Subsegment List Applet.

  > **Note:** Exceptions may arise if multiple variations of the same Business Component or Applet are needed. Typically, multiple variations will be necessary if a particular entity is displayed as both a top-level applet and a child applet on other views, and the two applets are not the same. In those cases, the name of the parent entity is appended to the beginning of the child applet name. For example, the ABC Account Contact List Applet is a Contact List that is displayed as the child of an Account. It needs the word Account to distinguish itself from the standard ABC Contact List Applet, which is a different applet.

- MVG applets should always be named <BusComp> Mvg Applet. (Please note the casing of Mvg.) Pick applets should always be named <BusComp> Pick Applet. Association applets should always be named <BusComp> Assoc Applet.
- A profile applet should always be named <Entity> Profile Applet, not <Entity> Profile Form Applet.

A new PickList object should always be named ABC PickList <entity>. Note that if the entity name itself has a prefix, it doesn't need to be repeated. For example, a PickList based on the MS Subsegment business component would be ABC PickList Subsegment rather than MS PickList MS Subsegment.

- When creating physical extension column:
  - Use an _ID suffix for foreign key columns (varchar 15) and a _CD suffix for List of Values domain fields (varchar 30). Do not use these suffixes for fields that do not meet these criteria.
  - Use a physical type of varchar 40 for phone number fields.
  - Remember that an extension column on a base table will automatically be given an X_ prefix by the Siebel application, so it is not necessary to add an additional prefix (for example, X_ABC_) to distinguish the columns from the standard Siebel column.
  - Limit column names to 18 characters, which is the Siebel standard. Using this standard will allow the Siebel application to use the column names as the basis for related column names without ever approaching the database limit for column names.
  - On MS SQL Server 7 or Oracle, always extend a base table in favor of using an _X extension table. Note that this was not always feasible on MS SQL Server 6.5 (and prior versions) or Sybase SQL Server due to limits associated with page size and number of columns per table.

The Golden Rule: Whenever in doubt regarding how to name an object, use the existing objects in the standard Siebel repository as your guide. For example, when creating a new Association applet, you would notice the <BusComp> Assoc Applet naming convention. Take the extra time to examine the standard objects and conform to their established naming conventions instead of cluttering the repository with poorly named objects.

Development Standards for Siebel Script Languages and Object Interfaces

Siebel scripting languages provide the ability to add powerful extensions to the existing declarative configuration capabilities. However, use of script may degrade maintainability, compromise upgrade options, and increase debugging time. The general rule is that all other possibilities should be exhausted before choosing a script to satisfy some functional requirement. The remainder of this section expands on that idea.

When considering the use of Siebel script languages or Object Interfaces, the following these next tenets will help you avoid writing superfluous script:

- During the initial high-level design, determine if the desired functionality is already part of Siebel or is a separate Siebel application module.
- The Siebel product suite is quite broad and a variety of business requirements are already implemented as a business component, business service, or separate product. Sometimes the solution may be only a small piece of functionality (visibility settings, user properties, search specs on links, specialized behavior of the base class) or it may be entire application modules (Forecasting, ePricer, eConfigurator). Either way, it makes the most sense to use standard Siebel features instead of building a custom solution.
- The future benefit realized from using existing product functionality outweighs the additional licensing costs that may be incurred for using additional Siebel modules. Even when initial development costs seem low, the longer term costs for maintaining, upgrading, and debugging code can add up quickly. Using or augmenting existing product behavior leaves the majority of these issues to the Siebel product development organization.
- Use existing business services for modular, generic behavior.
- There are over 300 pre-built business services that ship with Siebel, most of which can be used by script, Workflows, or external programs. These provide data transformation, communication, server requests, and various other utilitarian functions. Use of these business services either individually or together can help avoid duplicate development efforts.
- Use declarative configuration mechanisms whenever possible.

Configuration capabilities expand with each Siebel release. As a result, there are a number of ways to configure different types of functionality in Siebel Applications. Though they may not always be immediately obvious, most types of functional behavior are possible via declarative configuration.

The following table lists built-in techniques that are preferred over use of script, or other custom techniques, to fulfill functional requirements. The areas shown are the ones which are the most often erroneously implemented using script by implementers with less experienced with, and knowledge of, Siebel capabilities.

Technology	Documentation
User Properties	*Siebel Tools Reference, Volume 2, Appendix F*
Runtime Events	*Siebel Workflow Administration Guide, Chapter 4*
Workflow	*Siebel Workflow Administration Guide*

State Model	*Siebel Workflow Administration Guide, Chapter 17*
Visibility	*Siebel Tools Reference, Volume 1, Chapter 4*

These technologies may be use to perform a wide variety of tasks, including simple data validation, event-driven processing, data-driven read-only behavior, interaction with external systems, security, state transitions, etc.

Using the guidelines shown above will help to enforce standards that will positively influence the two factors that have the most significant impact on application quality and costs: maintenance and upgrades. Use of standard Siebel components and declarative configuration tools decrease the time and effort required to maintain the application. Also, since Siebel Workflow, State Model, User Properties, etc., are configured using graphical UI tools, their use requires the lowest level of technical knowledge. Also, since functionality created using these built-in capabilities is stored in a single place, their use supports a more comprehensive view of the overall functionality that has been implemented. Conversely, understanding the functionality that has been implemented using scripts is much more laborious as because developers must look at each object individually and read the script code to understand what has been implemented.

Declarative functionality is upgraded automatically when the application is upgraded. Scripts are not. After an upgrade is complete, all objects that have script must be tested to ensure functionality has not been lost and all non-trivial scripts must be reviewed to ensure they are compatible with application changes. Configuration mechanisms are rarely removed from the application, and as a result the majority of standard Siebel functionality upgrades without incident. If an upgrade will affect declarative functionality, it will be documented in the version release notes.

Templates and Tools

The following templates and tools specific for Siebel implementations are available to assist with this task:

Template File Name	Comments
DS-020_APPLICATION_EXTENSION_STRATEGY_SIEBEL.DOCX	MS WORD

DS.090 Design Data

This task is used to detail information for new or modified applets.

Approach

MAP DIRECTLY TO THE DATA OBJECT LAYER

The Business object layer refers to the field, business component, and business object layer. The data layer refers to the columns and tables.

Fields (Business Layer) map directly to Columns (Data Layer). A Business Component (Business Layer) maps directly to a Table (Data Layer). A ruling principle explains that there is one Business Component per applet and one Business Object per view. It is important to know that each applet maps to one and only one business component. However, many different applets can be created from the same business component.

A Business Component represents a business entity within the Siebel application such as Account, Contact, and Service Request. A business component can be thought of as a virtual database table that spans multiple real tables. It organizes the data in the way the user chooses to view the data and not how it is organized for effective data storage.

A Business Object represents a collection of related business components used in a major functional area of the business. The majority of business objects have one business component that serves as the parent or driving business component.

MAP THE BUSINESS OBJECT AND DATA LAYERS

To complete this critical portion of the Gap Analysis effort, the business analyst will work with the technical configurators to understand how fields map to columns in a table within the database. Working collaboratively with the configurator, the business analyst will designate relationships as many to one, many to many, one to many, or one to one. These parameters can be demonstrated as M:1, M:M, 1:M, or 1:1.

It is important to keep the following parameters in mind when completing this portion of the Gap Analysis:

- One field maps directly to a column and to one Business Component (BC).
- Fields have a many-to-one (M:1) relationship with BC.
- Columns map to one, and only one table giving it a many-to-one (M:1) relationship.
- BC map directly to tables.
- BC map to one and only one Business Object giving the relationship a many-to-one (M:1) relationship.

The mapping exercise will reinforce the ER Diagram explaining the 1:1, 1:M, M:1, M:M relationships. Although the Solution Design is not the Business Analyst's deliverable, it is important that the business analyst understand the layer structure so the architect and configurators will be better prepared for the upcoming solution design effort.

CREATE ENTITY RELATIONSHIP DIAGRAMS

The business analyst works closely with the technical configurators in defining the ER diagram. The diagram represents the relationship between the different entities or modules the customer has purchased.

Templates and Tools

The following templates and tools specific for Siebel implementations are available to assist with this task:

Template File Name	Comments
DS-090 APPLETS DESIGN WORKBOOK.XLS	MS EXCEL-This document only contains information for new or modified applets. In addition, for modified applets, only new or modified fields are included. Be sure to remove any yellow notes before presenting this document to the customer.
DS-090 BUSINESS COMPONENTS DESIGN WORKBOOK.XLS	MS EXCEL-This document only contains information for new or modified applets. In addition, for modified applets, only new or modified fields are included. Be sure to remove any yellow notes before presenting this document to the customer.
DS-090 INTEGRATION OBJECTS DESIGN WORKBOOK.XLS	MS EXCEL-This document only contains information for new or modified applets. In addition, for modified applets, only new or modified fields are included. Be sure to remove any yellow notes before presenting this document to the customer.
DS-090 SCREENS DESIGN WORKBOOK.XLS	MS EXCEL-This document only contains information for new or modified applets. In addition, for modified applets, only new or modified fields are included. Be sure to remove any yellow notes before presenting this document to the customer.
DS-090 VIEWS DESIGN WORKBOOK.XLS	MS EXCEL-This document only contains information for new or modified applets. In addition, for modified applets, only new or modified fields are included. Be sure to remove any yellow notes before presenting this document to the customer.
DS-090_DATA_DESIGN_SIEBEL.DOCX	MS WORD

DS.110 Design Business Rules

The following areas are involved during the business rules design in a Siebel implementation:

Design Workflow / Business Rules

Design the Workflow Manager Policies/Rules based on information gathered.

APPROACH

1. Design Workflow Manager policies/rules.
2. Prepare for Workflow Manager Design Review.
3. Design state models.
4. Design Assignment objects, rules.
5. Conduct Assignment Manager Design Review.

Design VB/COM Business Rules

Oracle recommends using great care in the use of Siebel VB/COM interfaces. Business rules or application extensions created in this way should:

- Focus on performance.
- Focus on maintainability.
- Properly use Siebel Objects and Methods.
- Optimize algorithms using Siebel Interfaces.
- Properly make use of configuration, Siebel VB, and external code (DLLs and executables) using Siebel COM interfaces.
- Follow common programming guidelines.

APPROACH

1. Explore declarative configuration alternatives.
2. Determine where to put the script: Which Object?
3. Determine where to put the script: Which Event?
4. Add error handling template.
5. Add method body.
6. Test.

Basic Process Flow

Explore Declarative Configuration Alternatives - Before writing any script, developers should explore all declarative configuration alternatives. This is an important step. It ensures that developers write script only when there is no configuration alternative, thus minimizing the amount of script.

Determine Where to Put the Script: Which Object? - After determining that the solution requires script, developers must determine where to put the script. A utility script that will be called from many locations such as applets, business components, and business services, is best implemented as a business service. This enables developers to write the script once for use by many objects. An alternative to writing a business service is to write the method at the application level. Oracle encourages developers to use business services, as they can be called by workflow processes. Workflow processes are not able to call custom application-level methods. If the method deals with data in a specific business component, that business component is the most likely object for the script. Writing script at the business component level prevents developers from having to re-write it on applets that leverage the same underlying business component. This only makes sense if the method is specific to a particular business component. If the same script is seen in many business components with only slight differences, move the script to a business service and parameterize it. If the method controls the behavior of an applet, for example, if it is enabling a button or hiding/showing list columns or controls, write the script at the applet server level. If the method facilitates interaction with the client's desktop or communication with the user, make the script an applet browser script.

Determine Where to Put the Script: Which Event? - After determining the appropriate object, developers must determine the appropriate event within that object. There are two ways to think of scripting a business component: proactively and reactively. If you need to perform some sort of validation, but do not need to change data in any record other than the current record, use the Pre- event (for example, PreQuery, PreSetFieldValue, PreWriteRecord). This is a proactive approach, as you are trying to stop or modify something the user is doing before it gets too far.

Add Error Handling Template - Put an error/exception handling strategy in place before writing any script. This includes creating an error handling template that you can apply to all scripted events. Applying the error handling skeleton to a method before writing the method, assures two things:

- Error handling is present.
- Future manipulation of the method will not cause a runtime error that goes unnoticed.

Add Method Body - Now you are ready to write the actual method. “Sandwich” this code within the error handling code, so that you can catch any runtime errors.

Test - This is the last and most important step of scripting. Make sure the tests are complete, covering all possible conditions. Failure to test completely results in more wasted time trying to track down a bug later.

Assignment Manager Rules

APPROACH

1. Review and setup object parameters.
2. Set up/review organization / position hierarchies.
3. Create assignment rules / criteria.

To produce outputs consistently, the techniques indicated in *Siebel Bookshelf: Siebel Assignment Manager Administration Guide* should be applied when completing this activity.

DS.130 Design User Interface

Templates and Tools

The following templates and tools specific for Siebel implementations are available to assist with this task:

Template File Name	Comments
DS-130_REPORTS_DESIGN_WORKBOOK_SIEBEL.XLS	MS EXCEL-This document only contains information for new or modified applets. In addition, for modified applets, only new or modified fields are included. Be sure to remove any yellow notes before presenting this document to the customer.
DS-130_USER_INTERFACE_DESIGN_SIEBEL.DOCX	MS WORD

DS.140 Prepare Design Specification

Where the characteristics of a project support it, the Design Specification (DS.140) may be used to consolidate a number of Design work products. The Design Specification can be used to create a single document that describes the design for each component of a custom extension.

The Design Specification work product should include or reference all the key object definitions to be changed in the repository and the key changes for each definition:

- Application, Screen, View, Applet
- Business Object, Business Component
- Database
- Pseudo Code with Objects and Events Identified
- Integration Objects and Other Interfaces

Physical and Logical Architecture
- ERD
- Data Load Mappings
- Administrative Data and Functions (Personalization, Product Definitions, Lists of Values, Workflow, Assignment Manager, etc.)

It is also worth mentioning here that while use cases can capture sufficient detail, proper analysis of existing data must be done and a clear design phase must still be undertaken before development begins. Data, in particular, may be a limiting factor in the ability to deliver functionality (that is data dependent or in the ability to deliver a usable system).

Templates and Tools

The following templates and tools specific for Siebel implementations are available to assist with this task:

Template File Name	Comments
DS-140_DESIGN_SPECIFICATION_SIEBEL.DOCX	MS WORD

DS.150 Develop Database Design

Check the *Expert Services Development Guidelines and Configuration Concepts* to review the guidelines related to Database Extensibility.

Database Extensibility

Database extensibility covers three main areas which are available as part of the standard product when using Siebel Tools.

- 'Static database extensibility" refers to the ability to adapt the database to suit an enterprise's specific needs without altering the schema in any way. This involves the use of predefined columns that have been reserved for this purpose. These columns can be in tables that have a one-to-one relationship with existing base tables, i.e. those predefined tables that have an 'S_' prefix and an '_X' suffix, or in tables that have a many-to-one relationship with an existing table, i.e. tables that have an 'S_' prefix and an '_XM' suffix.
- 'Dynamic database extensions' modify the schema in a well-defined way and allow new columns to be created on Siebel base tables. In addition, it may include the creation of new extension tables, whose records have a one-to-one relationship with records in a base table.
- 'Advanced database extensibility' provides the ability to create new tables whose records have a 1:M or M:1 relationship with records in a Siebel-supplied base table. An enterprise can create stand-alone tables that may or may not be related to Siebel base tables, as well as custom intersection tables in order to facilitate new many-to-many relationships, either with new stand-alone tables, or within the supplied schema.

Best practices dictate that an enterprise considers database extensions, they should consider them in the order listed above. Whenever possible, the supplied schema should be used. This provides for the automatic use of optimizations that are built in the standard Siebel application. If this is not possible, then the next preferred method is to create simple extensions to the supplied schema, adding columns using the Object List Editor, or creating extension tables through the use of the 'Extend' button, visible when viewing the 'Table' object within tools and with the ‘Newtable’ project checked out. The last option is to use advanced database extensibility features.

Siebel has addressed many of the issues related to Advanced Database Extensibility which has simplified the process of creating and managing complex extensions to the supplied database schema, while at the same time continuing to support mobile users and maintaining the ease of use with EIM. Siebel Tools contains a Docking Wizard and an EIM Wizard to automate the creation of the relevant records to support these features. The use of these wizards is clearly detailed in the *Tools Guide* within *Siebel Bookshelf*. You should be completely familiar with *Chapter 6*, Adding Custom Extensions to the Data Model, before attempting to extend the schema in any way. This chapter is updated with each release of Bookshelf so, even if you have read it for a prior version of Siebel, you should make yourself aware of the current options in subsequent releases. Many options exist as alternatives to creating custom tables and these should be exhausted before creating custom tables.

The use of advanced database extensibility adds risk to your projects and future upgrades. Therefore,it is highly recommended that you carefully consider your use of this feature. Once your plan is complete, you should contact your TAM (if you don't have one yet you should contact your salesperson) in order to schedule a Database

Extensibility Review with Siebel Expert Services (ACS – Advanced Custom Support). Expert Services have worked with the vast majority of Siebel customers and are therefore in a unique position to offer advice gained from all the mistakes that have been made in the past. Most of the issues that may occur when using advanced database extensibility are specific to a given situation, business need, and architecture; therefore it is not feasible to offer advice for all possible situations. Some of the things you will need to consider when planning modifications to the supplied schema are as follows:

- EIM support is not possible without assistance from Expert Services (ACS) for standalone tables.
- Certain extensions are not appropriate in Siebel due to fundamental changes to the schema. For example, it was previously possible to create a number of additional one-to-one extension tables for S_CONTACT, which is no longer possible as S_CONTACT is now an extension table of S_PARTY (it is not possible to create extension tables for existing extension tables).
- New tables will typically require new indexes. Index planning should be agreed between a qualified database administrator and Siebel Expert Services (ACS). Custom indexes can, even if correctly implemented, result in the standard Siebel indexes being ignored. This is likely to have an adverse impact on performance.
- While a '_XM' table is a logically sound extension to an existing '_XM' table, this will cause issues with EIM due to the design on the EIM processes. This should be avoided.
- Custom one-to-one extension tables will automatically support docking and mobile users; however, EIM mappings must still be created.
- There are some assumptions made in the creation of dock objects, when using the Docking Wizard, in order to specify appropriate visibility strengths. All custom foreign keys and docking rules created for them using the Docking Wizard should be reviewed by Expert Services (ACS).
- Modification of standard base tables and their columns is not supported.
- You should use upper case for any new tables or column names.
- It is also not possible to add EIM mappings for foreign key relationships to tables that do not have user key attributes.
- You must follow the steps set out in *Bookshelf* to support EIM and Docking in order to create a supportable Siebel system.

STATIC DATABASE EXTENSIONS

_X extension tables (1:1) - This type of extension table is basically just an extension of the base table record. You do not need to create a new business component object when you use these tables. The _X tables are implicitly joined to the parent table. For example, S_OPTY_X is implicitly joined to S_OPTY and is available in the Join picklist for a BC field without being explicitly defined in the BC.

_XM extension tables (1:M) - This type of extension table allows you to create a one to many relationship with a record in the parent Siebel base table. The following is a list of guidelines to follow when using the _XM tables:

- Create a new business component based on the relevant _XM table.
- Create a ‘Type’ field based on the TYPE column and use this to filter the records by specifying it in the search spec for the business component.
- Add the correct Pre Default Value to the Type field.
- Use the NAME column to track the main data field.
- Unique index is on PAR_ROW_ID, TYPE, NAME, CONFLICT_ID- you will need to consider this in your design.
- If displaying in MVGs, extend the base table to track primary values.

DYNAMIC AND ADVANCED DATABASE EXTENSIONS

As with all configuration work, database extensibility must first be performed against the local database. This provides recoverability in the event of mistakes and allows the developer to thoroughly test changes before making them available to other developers. The following steps detail the process:

1. Check Out the Project - With Siebel Tools running against the local database, Check Out the project that contains the table you need to modify. This includes the ‘Newtable’ project and may include other projects.
2. Update the Logical Schema Definition in the Local Environment. Make the desired changes to the tables, columns, indexes, and mappings contained in the ‘Newtable’ project. Refer to the Siebel Tools Reference Guide for further instructions on modifying the data model.
3. Apply the Physical Schema Extensions to the Local Database - Once your changes are complete, follow these steps to update your local environment.
    - Click the “Apply” button in the Table List window to apply the changes to your local database. You may receive a warning message that database extensibility from the client is not supported.
    - Specify Current Query, All or Current Row. Type your local database password (which must be in upper case), and click “Apply.”
    - Once this process has completed, press the “Activate” button in the Table List window.

    - Test the changes thoroughly. Typically, data model changes are exposed in screens or views. Test all of these objects, checking out updated copies modified by other developers, if necessary, against the local database.
4. Apply the Changes To The Server Database – Check in the “Newtable” project to update the repository schema definition on the server. At this point, the logical database schema of the server database has been updated, but the changes have not been applied to the physical server database.
5. Repeat Step 3 while connected to the server using Siebel Tools.

Ideally, you will make bulk changes to the Siebel schema at the outset of each project phase, thereby, minimizing the impact of changes on other developers. If changes are made during a project phase, then these will need to be distributed to all mobile users. It is possible to use Siebel Anywhere to distribute a schema change; otherwise, you must re-extract all mobile users prior to going live with the new phase. For Additional details, see the *Tools Guide* in the *Siebel Bookshelf*.

DS.160 Review Design Model

A configuration review with Siebel Expert Services (ACS) is always recommended. A configuration review is a code review of the customized application. The purpose of the review is to verify that best practices were followed during the customization, and that the configurations performed by the development team will work optimally. The review should be performed on a configuration that is completed and ready for deployment but prior to production rollout.

The files for the review need to be ready and uploaded 24 hours prior to the beginning of the review.

The major items checked during a design review are:

- visibility
- functionality
- clean and consistent user interface
- performance
- maintainability and upgradability
- assignment manager rules
- workflow rules and policies
- data transfer and conversion with EIM

Prepare for Configuration Design Review

Design reviews are conducted by a Configuration Specialist from Siebel Expert Services to ensure that data mappings, database extensions, and configuration designs meet the enterprise's stated functional requirements, maintain the integrity and performance of Siebel Enterprise Applications, and adequately use the configuration features available in Siebel Tools.

A Design Review is a review of the planned design for a Siebel implementation. The review includes an ERD review, data mapping review, review of applet designs, and information flow review. In the case of extensive or unusually complex Siebel VB usage, a Siebel VB structural design review is also recommended. The Design Review ensures that the enterprise is making optimal use of the Siebel data model and configuring the Siebel application according to recommended standards. A Design Review should be conducted prior to customizing the Siebel Repository for the implementation.

The input files for the review must be ready and uploaded 24 hours before the beginning of the review. The Configuration Team, assisted by the Technical Account Manager (TAM), if necessary, should notify the reviewer when the files have been uploaded, supplying their location and any other relevant details.

Prepare for VB/COM Design Review

In the case of extensive or unusually complex Siebel VB/COM usage, a Siebel VB/COM Design Review is recommended. The purpose of the Siebel VB/COM Design Review is to examine the resilience, efficiency, and clarity of the code written in the Siebel Application. As part of this review, the Siebel Expert Services review team does not assess whether or not the code accurately implements the user requirements of the design.

The Siebel VB/COM Design Review is a complete review of all VB code in the configuration that will support the Siebel implementation. This includes all Siebel VB code in

the repository and any Siebel-relevant parts of other VB code that may be accessed by the application. It is important that the reviewer be notified if any of the code resides outside of the Siebel repository.

The reviewer conducts the review with an eye to:

- Performance
- Design
- Maintainability
- Compactness
- Documentation
- Proper use of Siebel Objects and Methods
- Optimization of Algorithms using Siebel Interfaces
- Proper Delineation of Code between Configuration, Siebel VB and External Code (DLL's and executables) using Siebel COM Interfaces
- Adherence to Common Programming Guidelines

The files for the review need to be ready and uploaded at least 24 hours before the beginning of the review. This should include documents briefly describing the intent of the code and describing any preliminary questions from the enterprise.

Prepare for Workflow Design Review

The Workflow Design Review, conducted by Siebel Expert Services, is designed to address specific workflow design issues. Participants should include the business people who have asked for the policies and who know about business volumes that these policies govern, the technical team that has developed the workflow components (workflow objects, programs, actions, policies, groups), and the Siebel TAM, if possible. This review covers the following:

- Overall Workflow Objectives: Can workflow accomplish everything that the customer requires, or are there other means (Siebel Visual Basic, Assignment Manager, Pre-defaulting fields and so on) that should be considered?
- Custom Workflow Objectives: If non-vanilla Siebel Workflow Objects have been created, a review of the number of objects and the impact on the system.
- Workflow Programs: A review of the customer's workflow programs. Are they complete? Do they accomplish what is expected?
- Workflow Policies: A review of workflow policies to ensure their completeness and determine whether they can be further optimized. How do the policies and the data volumes affect the end users?
- Workflow Groups: Advice on how to split workflow policies into groups to achieve the customer's performance and throughput goals. This is one of the key elements in ensuring a successful implementation.
- The following deliverables are the outcome of the Workflow Design Review:
- A report that reviews all of the basic elements of the Siebel Workflow Manager and points out the strengths and weaknesses of the current design.
- An alert to any elements that are incomplete or do not meet the expectations of the customer, ensuring that these items are not discovered after system rollout.
- A set of strategies for grouping the policies to be most effective.

Prepare for Workflow Design Review

The Assignment Manager Review, conducted by Siebel Expert Services, is a pre-production review designed to ensure optimal design and implementation of Assignment Rules. The Assignment Manager Review may include the following, as appropriate to a particular customer implementation:

- Ensure that the business requirements are met via the Assignment Rules the enterprise has configured.
- Review any customizations made to Assignment Manager (for example, the addition of Assignment Objects or Assignment Criteria).
- Ensure Assignment Rules are optimally configured.
- Consider any performance implications of the Assignment Manager setup including a review of potential data volumes, triggers, etc.
- Resolve any specific Assignment Manager questions and issues.

An Assignment Manager Design Review returns a document containing suggestions and considerations for the enterprise.

Review Mappings with Application Design Team

The Configuration Team has the most complete understanding of how data is viewed and managed within the Siebel user interface. EIM data mapping should be reviewed with this team to ensure that converted and interfaced data appears in the user interface correctly and that all types of data are handled appropriately.

Prepare for EIM Mapping and Design Processing Review

The EIM Mapping Review is a review designed to ensure that the customer is using the correct EIM mappings to interface data into and out of Oracle Siebel Applications. In this review, Siebel Expert Services will examine the EIM mappings design and make sure that the enterprise is using the correct IF tables and mappings to import, update, delete, and merge data into the Siebel database. If desired, the enterprise can send their integration design document detailing the data integration processes across the various legacy systems for review. Areas that are reviewed are:

- Valid EIM Mappings
- Required Columns in Interface Tables for EIM Processing
- IFB Files (optional)
- Data Integration Design Document (optional)

The Architecture Specialist will review all mappings specified in the mapping spreadsheet for validity. If IFB files are provided, the Siebel Expert Services review team will check for syntax errors and processing order. The team also provides feedback on data integration design (across various legacy systems and Siebel products), if the design documents are provided.

A comprehensive EIM Mappings Review document that details findings and recommendations specific to the enterprise's environment will be returned to the enterprise after the review is completed, typically via the account’s TAM.

Review Data Mapping

As with other data mapping documentation, the information between the User Interface and the Business Object should be reviewed with the Configuration Team. A Business Analyst and an Expert Services Configuration Specialist would also be ideal to also crosscheck this specific mapping for accuracy.

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IMPLEMENTATION (IM) PROCESS GUIDANCE

IM.005 Develop Conceptual Prototype

Preparation of a Conceptual Prototype may be warranted when a visual representation of the requirements as currently understood would be helpful in confirming/validating the requirements. A high-level application prototype simulates some aspect of the functionality, design, structure, and operational characteristics of the system. It usually describes the required "look and feel" of the user interface, system business scope, system topology, and other factors that contribute to the vision. This prototype may be completed on-line, on paper, or with a combination of both.

IM.040 Implement Database

Create Customized Physical Database

Once the standard Siebel database has been installed and established, any customizations must be made before new UI and other features that require them are implemented.

The first step of configuration is to make any local database table or record changes. Any additional fields, customized tables 1:1 or M:1 extension tables, new or modified intersection tables or other database changes must be implemented.

1. Implement database extensions to user tables.
2. Create EIM IF table extensions and related mappings.

Database Extensibility

Database extensibility covers three main areas which are available as part of the standard product when using Siebel Tools.

- 'Static database extensibility" refers to the ability to adapt the database to suit an enterprise's specific needs without altering the schema in any way. This involves the use of predefined columns that have been reserved for this purpose. These columns can be in tables that have a one-to-one relationship with existing base tables, i.e. those predefined tables that have an 'S_' prefix and an '_X' suffix, or in tables that have a many-to-one relationship with an existing table, i.e. tables that have an 'S_' prefix and an '_XM' suffix.
- 'Dynamic database extensions' modify the schema in a well-defined way and allow new columns to be created on Siebel base tables. In addition, it may include the creation of new extension tables, whose records have a one-to-one relationship with records in a base table.
- 'Advanced database extensibility' provides the ability to create new tables whose records have a 1:M or M:1 relationship with records in a Siebel-supplied base table. An enterprise can create stand-alone tables that may or may not be related to Siebel base tables, as well as custom intersection tables in order to facilitate new many-to-many relationships, either with new stand-alone tables, or within the supplied schema.

Best practices dictate that an enterprise considers database extensions, they should consider them in the order listed above. Whenever possible, the supplied schema should be used. This provides for the automatic use of optimizations that are built in the standard Siebel application. If this is not possible, then the next preferred method is to create simple extensions to the supplied schema, adding columns using the Object List Editor, or creating extension tables through the use of the 'Extend' button, visible when viewing the 'Table' object within tools and with the ‘Newtable’ project checked out. The last option is to use advanced database extensibility features.

Siebel has addressed many of the issues related to Advanced Database Extensibility which has simplified the process of creating and managing complex extensions to the supplied database schema, while at the same time continuing to support mobile users and maintaining the ease of use with EIM. Siebel Tools contains a Docking Wizard and an EIM Wizard to automate the creation of the relevant records to support these features. The use of these wizards is clearly detailed in the *Tools Guide* within *Siebel Bookshelf*. You should be completely familiar with *Chapter 6*, Adding Custom Extensions to the Data Model, before attempting to extend the schema in any way. This chapter is updated with each release of Bookshelf so, even if you have read it for a prior version of Siebel, you should make yourself aware of the current options in subsequent releases. Many options exist as alternatives to creating custom tables and these should be exhausted before creating custom tables.

The use of advanced database extensibility adds risk to your projects and future upgrades. Therefore,it is highly recommended that you carefully consider your use of this feature. Once your plan is complete, you should contact your TAM (if you don't have one yet you should contact your salesperson) in order to schedule a Database Extensibility Review with Siebel Expert Services (ACS – Advanced Custom Support). Expert Services have worked with the vast majority of Siebel customers and are therefore in a unique position to offer advice gained from all the mistakes that have been made in the past. Most of the issues that may occur when using advanced database extensibility are specific to a given situation, business need, and architecture; therefore it is not feasible to offer advice for all possible situations. Some of the things you will need to consider when planning modifications to the supplied schema are as follows:

- EIM support is not possible without assistance from Expert Services (ACS) for standalone tables.
- Certain extensions are not appropriate in Siebel due to fundamental changes to the schema. For example, it was previously possible to create a number of additional one-to-one extension tables for S_CONTACT, which is no longer possible as S_CONTACT is now an extension table of S_PARTY (it is not possible to create extension tables for existing extension tables).
- New tables will typically require new indexes. Index planning should be agreed between a qualified database administrator and Siebel Expert Services (ACS). Custom indexes can, even if correctly implemented, result in the standard Siebel indexes being ignored. This is likely to have an adverse impact on performance.
- While a '_XM' table is a logically sound extension to an existing '_XM' table, this will cause issues with EIM due to the design on the EIM processes. This should be avoided.
- Custom one-to-one extension tables will automatically support docking and mobile users; however, EIM mappings must still be created.
- There are some assumptions made in the creation of dock objects, when using the Docking Wizard, in order to specify appropriate visibility strengths. All custom foreign keys and docking rules created for them using the Docking Wizard should be reviewed by Expert Services (ACS).
- Modification of standard base tables and their columns is not supported.
- You should use upper case for any new tables or column names.
- It is also not possible to add EIM mappings for foreign key relationships to tables that do not have user key attributes.
- You must follow the steps set out in *Bookshelf* to support EIM and Docking in order to create a supportable Siebel system.

STATIC DATABASE EXTENSIONS

_X extension tables (1:1) - This type of extension table is basically just an extension of the base table record. You do not need to create a new business component object when you use these tables. The _X tables are implicitly joined to the parent table. For example, S_OPTY_X is implicitly joined to S_OPTY and is available in the Join picklist for a BC field without being explicitly defined in the BC.

_XM extension tables (1:M) - This type of extension table allows you to create a one to many relationship with a record in the parent Siebel base table. The following is a list of guidelines to follow when using the _XM tables:

- Create a new business component based on the relevant _XM table.
- Create a 'Type' field based on the TYPE column and use this to filter the records by specifying it in the search spec for the business component.
- Add the correct Pre Default Value to the Type field.
- Use the NAME column to track the main data field.
- Unique index is on PAR_ROW_ID, TYPE, NAME, CONFLICT_ID- you will need to consider this in your design.
- If displaying in MVGs, extend the base table to track primary values.

DYNAMIC ADVANCED DATABASE EXTENSIONS

As with all configuration work, database extensibility must first be performed against the local database. This provides recoverability in the event of mistakes and allows the developer to thoroughly test changes before making them available to other developers. The following steps detail the process:

1. Check Out the Project - With Siebel Tools running against the local database, Check Out the project that contains the table you need to modify. This includes the 'Newtable' project and may include other projects.
2. Update the Logical Schema Definition in the Local Environment. Make the desired changes to the tables, columns, indexes, and mappings contained in the 'Newtable' project. Refer to the Siebel Tools Reference Guide for further instructions on modifying the data model.
3. Apply the Physical Schema Extensions to the Local Database - Once your changes are complete, follow these steps to update your local environment.
    - Click the "Apply" button in the Table List window to apply the changes to your local database. You may receive a warning message that database extensibility from the client is not supported.
    - Specify Current Query, All or Current Row. Type your local database password (which must be in upper case), and click "Apply."
    - Once this process has completed, press the "Activate" button in the Table List window.
    - Test the changes thoroughly. Typically, data model changes are exposed in screens or views. Test all of these objects, checking out updated copies modified by other developers, if necessary, against the local database.
4. Apply the Changes To The Server Database – Check in the "Newtable" project to update the repository schema definition on the server. At this point, the logical database schema of the server database has been updated, but the changes have not been applied to the physical server database.
5. Repeat Step 3 while connected to the server using Siebel Tools.

Ideally, you will make bulk changes to the Siebel schema at the outset of each project phase, thereby, minimizing the impact of changes on other developers. If changes are made during a project phase, then these will need to be distributed to all mobile users. It is possible to use Siebel Anywhere to distribute a schema change; otherwise, you must re-extract all mobile users prior to going live with the new phase. For additional details, see the *Tools Guide* in the *Siebel Bookshelf* and the *Expert Services Development Guidelines and Configuration Concepts*.

IM.060 Perform Component Review

Siebel Expert Services group (ACS) can be involved in this activity to perform reviews and audits of the implementation.

The Expert Services tasks should include the following activities:

1. EAI Design Review
2. Design Review
3. CTI Design Review

4. Workflow Policy Manager Design Review
5. EIM Data Mapping Review
6. Configuration and Script Review
7. EAI Configuration Review
8. Production Readiness Review
9. Production Tune and System Audit

IM.090 Create Installation Routines

Carefully planned projects should proceed step-by-step to build and test demonstrable versions of software from a source repository that is carefully managed. Once the project is ready to move from development environment into the testing environment, or to move to the production environment, it is necessary to migrate the appropriate parts of the system in whole or in part as needed. This may mean migrating the complete master repository, migrating some parts of the repository, migrating setup data, and/or migrating user/test data. It is important to think through what will be required and define a process to control these migrations at this point in the project in order to be sure a flawless delivery to the target environment.

Define Operational Process for Migrating Repository to Target Environment

Identify and document all steps in sequential order from beginning to end which are required to perform what is commonly called a "dev2prod" from Siebel development to the testing or production environment. This document should include all tasks necessary to prepare the target environment to receive the appropriate updates and for all new or existing services to be in proper end state in the target environment. The plan will become the template that can be used for all future environment migrations and will be a critical tool for the customer team to use as they deploy post-implementation updates to their Siebel solution. The dev2prod process and template should be tested in a dev to test environment migration before executing a migration to the production environment.

Define Process for Applying Data Conversions to Target Environment

Identify and document all data conversion tasks and data updates that must be performed in the target environment.

Define and Document Post Migration Smoke Testing Process for Target Environment

After all migratory tasks have been completed, a checklist - testing process should be executed to verify that the environment and application are production ready. A smoke test consists of verifying that key functionality and data in the Siebel eBusiness application are working as expected.

Related Documents

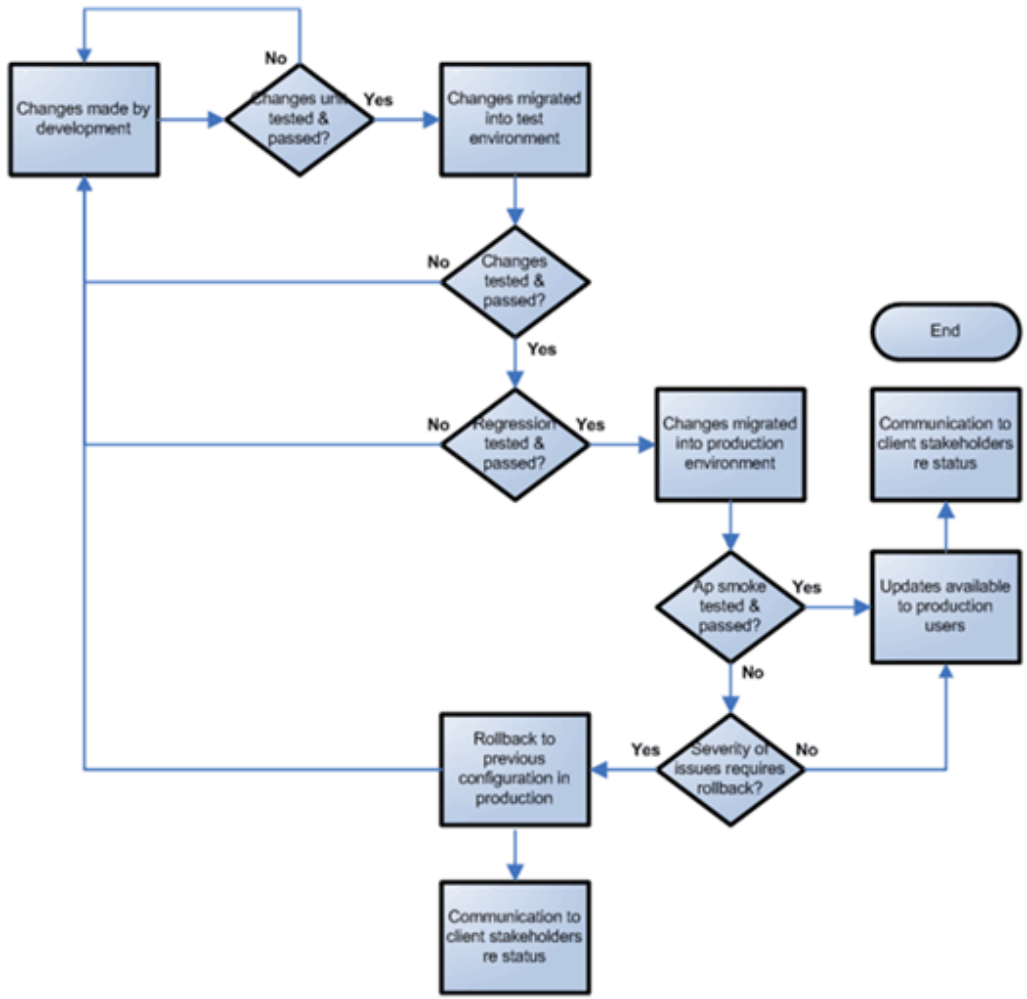
- Siebel Bookshelf: Going Live with Siebel eBusiness Applications
- Siebel Bookshelf: Siebel Remote and Replication Manager Administration Guide > Administering Siebel Remote > Sending Messages to Mobile Users
- Siebel Bookshelf: Applications Administration Guide: Message Broadcasting
- Siebel Bookshelf: Implementing Siebel eBusiness Applications> Siebel Repository Import, Export, and Postinstallation Tasks > Postinstallation Tasks for the Siebel Repository
- Siebel Bookshelf: Testing Siebel eBusiness Applications
- Siebel Bookshelf: Monitoring and Diagnostics Guide for Siebel eBusiness Applications

Executive Summary

Migrating changes, even small ones, to the Siebel production environment should be a repeatable process to minimize problems. Mitigating risks for a change to the production environment re-quires that specific conditions be satisfied before beginning the process and that post-update testing is performed before changes are made available to the business users and customers. An up-date to the production environment, commonly referred to as the dev2prod process must include:

- Implementation process that consistently follows Siebel’s best practices.

- Verification process that the changes to be implemented are defect free, that the portions of the application which have not been changed are stable, and that the production environment is in a steady state.
- Validation that the production environment has not been negatively impacted following up-date and that changes are working as expected.
- Rollback plan in the event that the in-progress implementation must be abandoned.



Production Smoke Test

A “smoke test” following migration of changes into the production environment is done to verify several key success elements for the release:

- The intended changes were migrated to the production environment
- New or updated functionality is working as expected
- Components of the application which were not changed are working as expected
- The application and environment are stable

A smoke test is a “spot check” of new configuration and of unchanged elements in the application after changes have been made to the production environment. It is unrealistic to expect that this test will be of the same scope as what was executed in preparation for the release. It is also not possible to test each function in the production environment. A post-implementation smoke test should not be considered a replacement for the thorough testing which is prerequisite to every release.

Smoke test scenarios are created by selecting key functions in the application that have been added or changed as well as those that have not been touched. Each Siebel implementation contains certain functionality and information that is critical to the customer's business processes. Creating test scenarios to verify that at least some of those mission critical functions in the Siebel solution are working as expected is prudent, regardless of whether changes were made to the configuration that supports those processes. Other candidates for inclusion in the smoke test effort are functions or data for which this release is correcting a defect.

If the implementation includes mobile users, dedicated client users, and/or wireless clients, there will need to be test scenarios specific to those types of client applications. Many of the test scenarios executed for the Siebel Web Client can also be repeated with the Siebel Mobile Client and/or Siebel Dedicated Client for additional validation and to prove that the application is working as expected across all client types.

Work Product Details

This document defines key information and documented process for migrating updates to the production environment. These releases may be maintenance or sustain releases which include small enhancements and/or changes to resolve defects in the production configuration or may be a release that consists of new functionality for the current Siebel version that has been installed in the production environment. Version upgrades of Siebel and/or RDBMS are out of scope of the release process described in this document.

The installation steps indicated in this document are created to ensure that production releases follow a consistent, reliable process, and that a plan for testing the changes in the production environment exists. This document contains information that is commonly required before the customer production release management team will approve a change to a production application. Much of the content in this document will be the same for subsequent releases in the customer's environment. After customizing the template as needed, the completed document can be reused as a template for deployment planning and delivery.

SCOPE OF INSTALLATION INSTRUCTIONS DOCUMENT

This template contains places to record the following information:

- Deployment team members and contact information
- Specific information about the hardware and infrastructure in the production environment
- Deployment checkpoints
- Entry criteria that must be met in order to begin deployment to production environment
- Exit criteria that must be met in order to conclude that deployment is successful
- Step-by-step plan for Siebel repository migration
- Rollback criteria which would cause deployment to be cancelled
- Post deployment smoke test plans
- Other changes in production environment that are coincidental to Siebel changes
- Status report contacts

AUDIENCE

Implementation team, customer stakeholders, and release managers for Siebel eBusiness Applications.

Templates and Tools

The following templates and tools specific for Siebel implementations are available to assist with this task:

Template File Name	Comments
IM-090_INSTALLATION_INSTRUCTIONS_SIEBEL.DOCX	MS WORD

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TESTING (TE) PROCESS GUIDANCE

TE.020 Develop Unit Test Scripts

Unit testing insures that a component within a module functions according to program specifications. Each component of a module should have its own test plan, separated by test objectives. These objectives should cover normal processing, error handling, and abnormal processing. A module component can also include an interface. Each interface between modules should also be unit tested, treating the areas beyond the exit and entry points of the interface as black boxes. Interface unit testing should verify that the inputs and outputs are in the accepted formats. Integration testing will verify each interface's results from start to finish between modules.

This activity will focus on only the changes/modifications from Vanilla Siebel. Vanilla Siebel areas that are unmodified are outside the scope of Siebel Unit Testing. Each Siebel developer will be responsible for both the planning and execution of entities/tabs/views they develop. Unit Testing will comprise of two general areas:

- Verify the functionality and behavior of new and modified applets, views, and screens against a standard checklist.
- Verify the applet and/or view specific extended functionality and behavior against a prepared unit test plan.

Each developer should first create a Unit Test Plan based on a standard checklist. The purpose of standard checklist is to cover the test areas shared by all applets, views, and screens. This checklist is preferred over listing every little detail to be verified for every applet, view, and screen. The latter approach would be an excessive task that would only jeopardize timely completion of a fully developed and unit tested application ready for system test.

The following high-level steps should be performed in this unit test process:

1. Test Plan for specific extended functionality and behavior. A template is provided bellow.
2. The specialized unit test plan needs to be reviewed prior to test execution for completeness.
3. The standard test checklist should be executed and verified.
4. The specialized unit test plan needs to be executed and verified.

The following form represents the standard checklist to be followed by each developer. This form should be completed and initialed for each applet, view, and screen.

Applet Testing

	Organization / System:	CRM Common Sales Platform Project	**Date:**		
	Applet:		**Entity:**		
Title:	***Siebel Applet Standard Test Verification***				

Number	**Functionality / Behavior**	**Verified (Initialed)**
	All Applets	
A1.	Applet title displayed in upper left corner of applet?	
A2.	Field display names consistent with design specification? Display names should capitalize the first letter of each word?	
A3.	Dropdown pick list arrow displayed and accesses appropriate pick applet?	
A4.	All editable number/integer fields have a dropdown calculator available? Non-editable number/integer fields do not have a dropdown calculator?	
A5.	All editable date fields have a calendar popup? Non-editable date fields do not have a calendar popup?	
A6.	All currency fields display the appropriate default currency symbol? Dropdown on currency field displays appropriate currency popup whether or not the field is editable? Currency popup provides ability to change the currency type if the currency field is editable?	

A7.	All fields that typically store more information than they display should have a detail popup (…) button? This should include both editable and non-editable fields.	
A8.	All Boolean (tick) fields should use a centered tick and not simply Y or N?	
A9.	All phone and fax fields allow proper entry and formatting of international and domestic phone numbers?	
A10.	If design specifications indicate that Inserts are allowed for this applet, verify that a new record can be successfully created? If Inserts are not allowed as noted in the design specifications, verify that you cannot create a new record in the applet? This action should be performed both locally and connected to the server.	
A11.	After creating a new record, make sure the record remains visible in the view on which it was created after leaving and returning to that view?	
A12.	If design specifications indicate that Deletes are allowed for this applet, verify that a record can be successfully deleted? If Deletes are not allowed as noted in the design specifications, verify that you cannot delete a record in the applet? This action should be performed both locally and connected to the server.	
A13.	When you delete a record, verify what should happen to the child records? Are they deleted? Are they not deleted? If they are not deleted, are they still accessible / viewable elsewhere in the application. If the child records of a deleted parent record should be deleted, verify they are actually deleted from the database and not simply no longer visible?	
A14.	If design specifications indicate that Updates are allowed for this applet, verify that an existing record can be updated? If Updates are not allowed as noted in the design specifications, verify that you cannot update any field in the applet? This action should be performed both locally and connected to the server.	
A15.	All fields that are specified as read only in the design specification cannot be updated for both new and existing records. All fields that are not read only as indicated in the design specifications can be modified.	
	List Applets	
A16.	*Default* order of columns matches design specifications?	
A17.	*Default* spacing/size of columns is reasonable for the data it represents.	
A18.	All date and alphanumeric fields, including alphanumeric fields that store principally numeric values should be left justified? All currency and numeric fields should be right justified?	
A19.	Data sorted appropriately as specified and verified in the business component sort specification? Default/predefined queries?	
A20.	Drilldowns / Threads: All hyperlinks/drilldowns are available and navigate to the appropriate view? Default drilldown on the far left of the list applet navigates to the appropriate view? Thread bar functionality works properly (e.g., navigates back to the original view) when used in conjunction with drilldown functionality?	
A21.	Verify if an alpha tab is or is not present in accordance with the design specifications? Alpha tab (if available) functions properly and is associated with the appropriate field?	
	Form Applets	
A22.	New, Delete, Copy, and Cancel buttons are present on the applet only if these actions are allowed? Read only form applets should not have these buttons present? If you can create, delete, and/or copy records in this form applet, then the New, Delete, and Copy buttons should be present on the applet?	
A23.	Verify visual presentation of applet in 800 x 600 resolution. No text is cut off or missing?	
A24.	Verify visual presentation of applet in 1024 x 768 resolution. No text is cut off or missing?	
A25.	All text boxes and controls are aligned horizontally? Aligned vertically? Spaced equally?	
A26.	All fields, including currency and numeric fields should be left justified?	
A27.	All tab orders are correct.	

	Chart Applets	
A28.	Verify that the Chart Axes are correctly labeled, and that they reflect the design.	
A29.	Verify that the Chart Views can be expanded where applicable.	
A30.	Verify that the drop-down menus are correctly set up, are active, and have a complete set of values as per the design.	
A31.	Verify the visual layout of the individual charts on each applet.	
A32.	Verify that by double-clicking on an individual component of a chart, it behaves as if a query is being done on that specific component.	
A33.	Verify that the axis value can be changed where appropriate, and that the corresponding view is presented.	

View Testing

	Organization / System:	CRM Common Sales Platform Project	**Date:**		
	Applet:		**Entity:**		
Title:	***Siebel Applet Standard Test Verification***				

Number	**Functionality / Behavior**	**Verified (Initialed)**
V1.	The view name in the navigation bar is present and matches design specifications?	
V2.	The view name is in the screen menu and matches design specifications and match the view name in the navigation bar?	
V3.	The view name is in the screen menu and matches design specifications and match the view name in the navigation bar?	
V4.	The view name in the title bar is present and matches design	
V5.	The view is appropriately distributed when viewed with eight sectors?	
V6.	The view is appropriately distributed when viewed with six sectors?	

Templates and Tools

The following templates and tools specific for Siebel implementations are available to assist with this task:

Template File Name	Comments
TE-020_UNIT_TEST_SCENARIOS_SIEBEL.DOCX	MS WORD

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PERFORMANCE MANAGEMENT (PT) PROCESS GUIDANCE

PT.010 Conduct Performance Management Workshop

This task is very important since performance is key for the users of an application in terms of:

- log in time

- log off time
- query entities instances en screens
- screen navigation times
- documents attaching for big size times, etc.

You should review the supplemental guide information for the task, RD.015, Determine KPI Collection and Reporting Strategy before accomplishing this task.

Templates and Tools

The following templates and tools specific for Siebel implementations are available to assist with this task:

Template File Name	Comments
PT-010 TECHNOLOGY_COMPETENCY_PERFORMANCE_TUNING_SERIES - EAI.DOCX	MS WORD

PT.050 Design Performance Test Scripts and Programs

Performance Testing provides an assessment of whether or not an infrastructure will perform and scale to company requirements both initially and based on known parameters of future growth. This task requires an image of the full database and all interfaces into the Siebel application (i.e., CTI, Middleware, email, etc.). Testing results impact the entire life cycle of the application since they identify whether the application performs to the specifications and whether or not additional hardware is required to support scalability. Through performance testing, Implementers establish a benchmark of the infrastructure performance, and this benchmark helps to proactively identify performance issues as the Siebel solution is installed in the enterprise. Benchmarks continue throughout the implementation process as new functionality is added, to help determine the scalability of the system and to analyze the impact of different numbers of users. This testing rigor verifies that the system can meet the current and future requirements of the enterprise.

The Performance Testing effort is an integral part of the validation process that begins when configuration of the Siebel application initiates and continues through deployment. Once the full range of functional testing is complete, a rigorous Performance Testing effort using system tools should be completed. Performance testing provides an assessment of whether or not the technology infrastructure will perform and scale, as required, in a deployed environment.

The use of automated tools is recommended to support conducting the three types of Load & Performance testing.

- Performance Test - determine reliability and stability of the application with anticipated load.
- Capacity Test - determine breaking point of an application.
- Longevity Test - a test that is performed over an extended period of time to determine durability of an application as well as capture any defects that manifests themselves over time.

The first step in this process is to establish a benchmark of performance through the completion of a performance test. Next, it is recommended that a capacity test be completed adding users until reaching the number of users expected to use the system over the life of the application. Finally, a longevity test should be performed over an extended period of time to determine durability of an application as well as capture any defects that become visible over time.

In addition, a Performance Tuning Audit may be appropriate to ensure optimal performance across the Siebel enterprise architecture environment. During this review, Expert Services (ACS: Advanced Customer Services) will proactively identify any performance issues, thus reducing the risks of issues at runtime and increasing confidence as the implementation goes live.

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TECHNICAL ARCHITECTURE (TA) PROCESS GUIDANCE

TA.010 Conduct Technical Architecture Workshop

This task requires the key participation of Siebel Expert Services group (ACS: Advanced Customer Services), so, some expert services should be involved in this workshop.

TA.030 Define Integration Requirements and Strategy

There are several aspects of developing an integration architecture that should be considered as you work through the integration strategy process. The first component is defining a data strategy. The areas of consideration are the requirements for data ownership, integration constraints, and data entry points to create a customized data strategy solution.

In addition to defining the data strategy, it is important to do a full analysis of integration requirements to identify the impact on front end and back end systems. The data strategy and integration requirements will be direct inputs into the development of the Integration Strategy, which defines the architecture and technology for the transfer and flow of data between impacted systems.

The purpose of this task is to build a consensus agreement on how the integration needs of the business should be addressed for a particular system. Each system that is under consideration for integration to the Siebel solution should be reviewed by subject matter experts from the customer's IT organization as well as the integration and technical architects from the project team.

1. Review the business functions provided by the system that Siebel will need to access through integration along with the expectations from the business (including frequency, performance, and scalability of the interface).
   These functions should also be cross-referenced to each of the processes or use cases that require the functions being provided by the system.
   Note: This will require alignment to the business analysis being conducted on a separate track; understand the implications for timing of this workshop before it is scheduled.
2. Review the types of messages that must be exchanged between Siebel and the target system in order to support the high-level business needs, making note of any issues identified.
3. Conduct a review of the integration options provided by Siebel and identify options that are suitable for the target system.
4. Identify platform, hardware, and networking issues that need to be addressed.
5. Identify any other issues and dependencies that will impact the successful design, development, and testing of the system interface.

TA.150 Define Final Platform and Network Architecture

Templates and Tools

The following templates and tools specific for Siebel implementations are available to assist with this task:

Template File Name	Comments
TA-150_FINAL_PLATFORM_AND_NETWORK_ARCHITECTURE_SIEBEL.DOCX	MS WORD

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DATA ACQUISITION AND CONVERSION (CV) PROCESS GUIDANCE

CV.020 Define Data Acquisition, Conversion and Data Quality Strategy

The work product also should describe the requirements for accessing existing data via the new system. The data to be made available is identified and the integrity of the data is examined. Requirements include:

- a mapping of the existing data required to the Conceptual Data Model entities
- specific access requirements for each entity
- data integrity

CV.025 Define Data Acquisition and Conversion Standards

Conversion Design

Data from legacy systems is often needed in the new Siebel system for reference or for ongoing operation. Some data will be converted; that is, it will be moved from another system into the Siebel database, which will become the system of ownership and management for that data. Other data will continue to be managed and owned by other systems, but will be regularly synchronized with the Siebel database through an interface.

A design specification for each of these conversion components necessary to support the transfer of data from the legacy system(s) should include the following, as appropriate:

- a mapping of the legacy system(s) data elements to the new elements, including code transformations or edit rules
- the design, including parameters and language of implementation, for conversion components
- the dependencies and timing of execution of the suite of modules necessary to convert in support of the release plan, e.g., a region, a pilot office
- EIM processing design

To create an optimized plan to facilitate a smooth data conversion, the following steps should be taken into account:

1. Create Data Conversion Plan.
2. Define Data Cleansing Plan.
3. Design Initial Data Load.
4. Document EIM Mappings.
5. Review Mappings with Application Design Team.

Manual Data Conversion Procedures

Manual procedures required to prepare existing data for conversion to the new system. The procedures cover preparatory work required to capture or transform data, and correct or prevent integrity problems.

Automated Data Conversion Procedures

A description of the data to be converted to the new system, through an automated process, and process specifications for the software required. The following are included:

- data to be converted and conversion procedures. Includes matrices showing mapping of old database attributes to new database attributes
- conversion steps

CV.040 Design Conversion Components (Initial Load)

There are two items that must be completed before initiating this task. These items are:

- Assisted by the Siebel Technical Account Manager (TAM) and the system integrator, the enterprise should provide a completed EIM Mappings spreadsheet which must include:
  - Columns or fields that are to be transferred from the legacy system into the Siebel application or vice versa. These should include data types and lengths and required values if any.
  - Siebel application business component fields and database table columns that will store the data to be loaded or extracted from/to the legacy system. These should also include data types and lengths and required values if any.
  - Interface tables and Interface columns to be used for EIM processing the data transfer between the legacy system and the Siebel application. Information on any default values should be included here if needed.
- Optionally, EIM configuration IFB files and process design documents on the integration strategy can be provided for the review.

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TRANSITION (TS) PROCESS GUIDANCE

TS.030 Develop Installation Plan

Templates and Tools

The following templates and tools specific for Siebel implementations are available to assist with this task:

Template File Name	Comments
TS-030_INSTALLATION_PLAN_SIEBEL.DOCX	MS WORD

TS.050 Prepare Production Environment
TS.052 Implement Production Support Infrastructure
TS.060 Go Production

To accomplish these activities refer to the Siebel Deployment Planning Guide and the Siebel Global Deployment Guide.

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DEFINITIONS AND TERMS

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A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | XYZ

A

Account
An organization or company that is a current or potential customer.

Account Briefing
See ACTIVE LANGUAGE.

ACD
See AUTOMATED CALL DISTRIBUTION.

Action
An event caused by the execution of an SQL statement, an external program, or both.

Active Language
The active language in the client's user interface when starting a Siebel application, as used and defined by the Language parameter in the .cfg file or by the /L command-line parameter.

Actuate
A report writer from the Actuate Software Corporation. The report writer consists of a viewer tool and a report design tool (the Actuate Developer's Workbench), which are integrated into Siebel applications.

Advisory Services
A term that refers to acquisitions and mergers in Siebel Finance.

Agent
A call center representative who places outbound calls, answers inbound calls, or both — such as for sales or support purposes.

ANI
See AUTOMATIC NUMBER IDENTIFICATION.

API
See APPLICATION PROGRAMMING INTERFACE.

Applet
1. In Siebel applications, a visual application unit that appears on screen as part of a view. 2. In Internet applications, a self-contained program unit. See also APPLET OBJECT TYPE.

Applet Designer
A visual design editor in Siebel Tools that allows developers to create or modify Siebel applets interactively. It is used to modify and extend list, form, tree, and chart applets, and the controls and list columns on the applets.

Applet Object Type
A Siebel object type that implements a visual application unit (an applet) that appears on screen as part of a view. An applet serves as the vehicle for data entry and navigation through the data, and may implement a data entry form, scrolling list table, chart, dialog box, navigation tree, or Web browser. It provides viewing, entry, modification, or navigation capabilities for data in one business component. See also BUSINESS COMPONENT OBJECT TYPE and VIEW OBJECT TYPE.

Application
A program or group of programs designed for end users. In Siebel products, an application is a subset of the Siebel Applications that can be invoked from an icon on the desktop or Start menu; for example, Siebel Sales or Siebel Service. See also APPLICATION OBJECT TYPE.

Application Administrator
People responsible for planning, setting up, and maintaining their company's Siebel applications. See also DATABASE ADMINISTRATOR and SYSTEM ADMINISTRATOR.

ApplicatioN Object Type
A Siebel object type. Each application is a defined subset of the constructs available in Siebel Applications. An application presents a grouping of screen object definitions.

Application Programming Interface
An API is a set of routines used by an application program to direct the performance of procedures by the computer's operating system.

Application Upgrader
A utility within Siebel Tools that allows developers to merge a repository (database of object definitions) from an older version of Siebel Applications, including one that a developer may have significantly customized, with a newer version. See also REPOSITORY.

Argument
1. An independent variable, used with an operator or passed to a subprogram that uses the argument to carry out specific operations. 2. An argument can also be an option to a command, in which case it is often called a command-line argument.

Association Applet
This applet provides the user with the ability to associate records of two business components that have a many-to-many relationship. An association applet is invoked from the Edit Add New Record menu option in a master-detail view.

Association Table
A synonym for intersection table. See INTERSECTION TABLE.

Asynchronous
Not synchronized; that is, not occurring at predetermined or regular intervals. This term is usually used to describe communications in which data can be transmitted intermittently rather than in a steady stream or when an immediate response is not necessary or possible.

Attachment Applet
See FILE ATTACHMENT APPLET.

Attribute
Sometimes used as a synonym for property. See PROPERTY.

Authentication
The process of identifying an individual, usually based on a username and password. Authentication merely ensures that the individual is who he or she claims to be, but says nothing about the access rights of the individual.

Automated Call Distribution
Telephony hardware (such as a switch) and software that work together to route calls to call queues and call center agents.

Automatic Number Identification

Technology that passes a caller's phone number along with the call. Through database lookup, for example, a caller or the caller's company can be identified as a Siebel customer.

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Base Column
A column in a database table in Siebel Tools that is used in Siebel Applications. The base columns contain descriptive information for primary or foreign keys that are defined and used by Siebel Applications. A base column is not necessarily a column in a base table.

Base Table
1. A table containing columns that are defined and used by Siebel applications. 2. A table that provides the essential columns to a business component; specified in the Table property of the business component. See also EXTENSION TABLE and TABLE OBJECT TYPE.

Basic Calls Window
A window that displays a list of Siebel VB subroutine and function calls that were executed prior to the current line. It is found in the Siebel VB Debugger. See also Siebel VB DEBUGGER.

Bookmark
A feature that uses Microsoft Word's internal field capabilities. A bookmark is used as a reference point within a Word document. For example, within Siebel Proposal Generator, you can map a Business Component field to a Word bookmark. The bookmark name in a briefing component must match the bookmark name referenced within Siebel exactly. It can only reference the current record within a business component. It cannot sum a series of records, and it cannot combine information from more than one field.

Boolean
1. A data type in Siebel Tools for a field or column where the values for that field or column can be either TRUE or FALSE. Developers specify a data type of DTYPE_BOOL to define a field as Boolean. 2. In the Siebel Object Types Reference , Boolean is one of the values that may appear in the Required column in the list of properties for an object type. A value of Boolean for a property, as opposed to Required, Optional, or System, indicates that it is a TRUE/FALSE property that will retain its default value if the developer does not specify TRUE or FALSE. Also called logicaloperator. See also OPERATORS and RELATIONAL OPERATORS.

Boolean Operators
Boolean operators enable you to test the truth or falsity of conditions. See also operators and relational operators. Briefing Dynamic reports that Siebel eBriefings creates by integrating information from multiple internal and external sources to give users targeted, up-to date business intelligence. A standard installation of Siebel eBriefings includes My Briefings, Account Briefings, Competitor Briefings, and Company Briefings.

Bulletins
Company broadcast messages published in My Briefings to alert all users, or users associated with specific positions and divisions, to internal company announcements.

BusComp
An abbreviation for business component, and the name of the object variable used to reference an instance of a business component in Siebel VB and the Siebel Object Interfaces.

Business Component Object Type
A Siebel object type that defines a logical representation of columns in one or more database tables. A business component collects columns from the business component's base table, its extension tables, and its joined tables into a single structure. Business components provide a layer of abstraction over tables. Applets in Siebel Applications reference business components; they do not directly reference the underlying tables.

Business Object
See BUSINESS OBJECT DEFINITIONS and BUSINESS OBJECT OBJECT TYPE.

Business Object Component Object Type
A Siebel object type that is used to include a business component in a business object. It is a child of the object type called Business Object.

Business Object Definitions
Object definitions that reside in the Business Objects layer of Siebel Applications. They are built on data object definitions and selectively combine and associate data object definitions into logical data constructs that are useful for application design. Some of these logical constructs are business components, fields, links, multi-value links, and business objects. See also BUSINESS OBJECT OBJECT TYPE and BUSINESS OBJECTS LAYER.

Business Object Interfaces

See SIEBEL OBJECT INTEFACES.

Business Object Manager
A synonym for Business Objects Layer. See BUSINESS OBJECTS LAYER.

Business Object Object Type
A Siebel object type that creates a logical business model using links to tie together a set of interrelated business components. The links provide the one-to-many relationships that govern how the business components interrelate in this business object.

Business Objects Layer
The architectural layer in the repository that holds business object definitions in Siebel Applications. The object definitions represent database data more abstractly than the object definitions in the Data Objects Layer, and define interrelationships between data object definitions. For example, some object types in this layer are Business Object, Business Component, Link, Field, Join, and Pick Map. See also BUSINESS OBJECT MANAGER, DATA OBJECTS LAYER and USER INTERFACE OBJECTS LAYER.

Button
A control with a Type property value of PushButton. It initiates an action when clicked.

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CA
See CERTIFICATE AUTHORITY.

Call Script
In Siebel SmartScript, an ordered set of questions and the transitions between them as defined by a Siebel applications administrator and used by a call center agent. A question represents a single interaction between a call script and the call center agent. An answer is a specific value defined for a question.

Cascading Menu
A submenu. Also called a hierarchical menu or child menu.

Case Sensitive
A program's ability to distinguish between uppercase (capital) and lowercase (small) letters. Programs that distinguish between uppercase and lowercase are said to be case sensitive.

Catalog
The hierarchy of products and categories in a model. See also CATEGORY, PRODUCT and MODEL.

Category
A group of related items in Siebel Product Configurator. A category can include any combination of products and other categories. Categories are defined in the Configuration Designer screen to organize the end user’s configuration choices in Siebel Quote Assistant. See also PRODUCT CONFIGURATOR.

Certificate Authority
A trusted third-party organization or company that issues digital certificates used to create digital signatures and public-private key pairs. The role of the CA in this process is to guarantee that the individual granted the unique certificate is, in fact, who he or she claims to be.

Channel Changer
The navigational aid that is visible in Siebel InterActive when accessed from a standard Web browser over the Internet. The Channel Changer consists of a number of buttons. Each button accesses a Siebel InterActive view or tool, such as an ActiveBriefing view or the Tracking Profile tool.

Channel Partner Company
A company that sells another company's products, such as a retailer, a distributor or a value-added reseller.

Chart
A graphic or diagram that displays data or the relationships between sets of data in pictorial rather than numeric form. See also CHART APPLET and CHART OBJECT TYPE.

Chart Applet

A style of applet that displays a business graphic. It is dynamically populated with data from the applet's associated business component.

Chart Object Type
A Siebel object type that implements a business graphic, such as a line graph, bar graph, or pie chart, in a chart applet. Properties in a Chart object definition control the acquisition, conversion, and presentation of data from the business component. Chart is a child object type of the Applet object type.

Check Box Control
A control in a form applet consisting of a small square into which an "Xmark" can be entered or cleared by clicking. It represents TRUE/FALSE data. A check box is implemented as a control with a value of CheckBox in the Type property.

Check In (v), Check-In (n)
In Siebel Tools, to move the object definitions in one or more projects from a developer's client-side repository to the server repository. A check-in also clears the server lock on the checked-in object definitions, making these object definitions available for use by other developers.

Check Out (v), Check-Out (n)
In Siebel Tools, to move the objects in one or more projects in the server repository to a developer's client-side repository. Check-out also places a lock on the checked-out object definitions, making these object definitions unavailable to other developers.

Child
The hierarchically subordinate member of a special relationship between two object types or object definitions: the parent-child relationship. A child object type has exactly one parent object type; a parent object type may have one or more child object types. For example, Site is a Field object definition that is a child of the Business Component object definition called Account. Similarly, a child object definition has exactly one parent object definition. For example, the Site Field object definition is a child object definition of the Account Business Component (parent) object definition.

Class
In object-oriented programming, a generalized category that describes a group of more specific items, called objects, that can exist within it. A class is a descriptive tool used in a program to define a set of attributes or a set of services (actions available to other parts of the program) that characterize any member (object) of the class.

Class Property
A property in Siebel Tools that is used in certain object types, such as Applet and Business Component. Selecting a Class property value assigns a style to the object definition, which dictates behavior and look and feel. For example, a value of CSSFrameList in the Class property in an Applet object definition makes it a list applet. There is no strict correspondence between class in the object-oriented programming sense and the Class property in an object definition in Siebel Tools.

Client
A computer that uses the services of other computers and accesses the data provided by other computers. Siebel applications support three types of clients, dedicated, mobile, and thin. See DEDICATED CLIENT, MOBILE CLIENT and THIN CLIENT.

Client/Server Architecture
A network architecture in which each computer on the network is either a client or a server.

Client Wins
A rule used to resolve a data conflict in Siebel Remote during a data merge, and Enterprise Integration Manager (EIM) and repository merges. During the merges, the data value from the client database takes precedence over the data value on the server database. To ensure consistent results, both client and server must employ the same rule. See also ENTERPRISE INTEGRATION MANAGER and SERVER WINS.

Cluster Services
A server platform that allows you to run Siebel Servers and the Gateway Server on two-node hardware clusters for high levels of availability and fault tolerance. See also CLIENT/SERVER ARCHITECTURE.

Column
A relational database structure that specifies a particular item of information to include in each row in a database table. It also specifies data type and length. See also COLUMN OBJECT TYPE.

Column Object Type
A Siebel object type that maps to an underlying database table column. Column is a child object type of the Table object type.

Combo Box
1. In Windows terminology, a control that combines a text box and an interdependent list box. 2. In Siebel applications, combo box functionality is usually implemented by using a text box

control that is configured as a static picklist. However, there is also a combo box control used primarily in chart applets. See also STATIC PICKLIST.

Combo Chart
A type of chart applet that displays a line graph overlaid on a bar chart.

Command
1. An instruction to a computer program that, when issued by the user or a program, causes an action to be carried out. 2. A command in Siebel CTI is an instruction, such as to make a call, that is sent from the Siebel client to the CTI middleware, for execution on the telephone switch.

Command Object Type
A Siebel object type that specifies the method invoked when a toolbar button associated with this command is executed. It also defines which bitmap appears on the toolbar button.

Common Object Request Broker Architecture
Also called CORBA, which is an architecture that enables pieces of programs, called objects, to communicate with one another regardless of what programming language they were written in or what operating system they are running on.

Component
A small binary object or program that performs a specific function and is designed in such a way as to easily operate with other components and applications.

Compound Query
In Siebel Tools, a selective search for object definitions in the Object List Editor that combines two or more search conditions.

Computer Telephony Integration
CTI is the means by which a computer system can interact with the telephone system. Telephony activity, such as making a call or transferring a call, can be initiated within a computer program such as Siebel Applications. Information about calls, such as ANI, DNIS, or call status data, is passed from the switch to the computer system. With CTI, the computer system supplements and extends the capabilities of the telephone switch—for example, by helping to route incoming calls to the most appropriate agent, or by presenting relevant customer records to agents taking calls. Siebel CTI works with other components to provide CTI capabilities for users of Siebel Applications.

Configuration
Typically in Siebel Tools, the modification of Siebel applications to change the look, feel, and behavior to meet specific business processes.

Configuration File
1. A Siebel file (extension .cfg) through which application configuration data is passed to the Siebel client program. 2. An automatically generated file containing data about the model. The name of this file is entered when a model is created. Configuration files are required by Siebel Quote Assistant and the Model Validation view.

Conflict Detection
The process of identifying inconsistent database change transactions between intermittently connected databases. For example, two updates with differing values to the same field of a table would cause a conflict. Siebel Remote detects duplicate conflicts, operation conflicts, and update conflicts. See also CLIENT WINS, MOBILE CLIENT, SERVER WINS and UPDATE CONFLICT.

Conflict Resolution
The process of applying a rule to conflicting transactions to determine which transaction has precedence for updating the database. See also Client Wins, Server Wins, and update conflict.

Control
A visual object displayed in a graphical user interface. It provides user interaction with the application. See also CONTROL OBJECT TYPE.

Control Object Type
A Siebel object type that implements a data control, typically in a form applet, with which the user can interact. Examples include text boxes, check boxes, and buttons. Generally, a Siebel control exposes data from one field in the business component or invokes programming logic (in the case of a button control). See also FORM APPLET.

Control Toolbar
In Siebel Tools, a toolbar that becomes active when the developer displays the Applet Designer. The buttons in the Control toolbar correspond to controls that can be added to the applet in the Applet Designer. See also A visual design editor in Siebel Tools that allows developers to create or modify Siebel applets interactively. It is used to modify and extend list, form, tree, and chart applets, and the controls and list columns on the applets.

CORBA
See COMMON OBJECT REQUEST BROKER ARCHITECTURE.

CTI
See COMPUTER TELEPHONY INTEGRATION.

Customization
A term that refers to the modified code of underlying object definitions to change the look and feel of an application. Siebel source code is not distributed as part of the product, and thus, cannot be customized. However, some customization is possible using Siebel VB. See also configuration and Siebel VB.

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Database
A collection of logically related data designed to meet a business need.

Database Administrator
An individual whose duties might include responsibility for the design, management, and implementation of the database. See also APPLICATION ADMINISTRATOR and SYSTEM ADMINISTRATOR.

Database Extension Designer
A utility in Siebel Tools that allows the developer to add columns and indexes to tables in Siebel Applications, or add new tables where extensions are required beyond the database tables as delivered.

Database Extract
The database extract component of Siebel Remote and Replication Manager that creates a database snapshot file for a given mobile user. The snapshot contains all of the data required to initialize the user's local database. A database extract retrieves data according to predefined routing rules.

Database Initialization
The process used to import data into an empty local database.

Database Management System
A software interface between the database and the user.

Database Server
The computer on which a shared database resides; the combination of hardware and software that enables users to access information stored in the database.

Data Management or Data Manager Layer
A synonym for Data Objects Layer. See also DATA OBJECTS LAYER.

Data Manager
A synonym for Data Objects Layer. See also DATA OBJECTS LAYER.

Data Mart
Data marts, which focus on a particular subject or department, are subsets of data warehouses. The larger data warehouse combines databases across an entire enterprise.

Data Model
See SIEBEL DATA MODEL.

Data Object Definitions
Object definitions that reside in the Data Objects layer. They directly map the data structures from the underlying relational database into Siebel Applications, providing access to those structures by object definitions in the Business Objects layer. Data object definitions create a layer of abstraction over the DBMS, insulating the application and the developer from database administration and restructuring.

Data Objects Layer
The architectural layer in the repository that consists of data object definitions. These include object definitions that directly represent tables, columns, and indexes in the underlying DBMS. They include the Table, Column, Index, and Index Column object types.

Data Security
A scheme for protecting data from intentional or unintentional update or deletion. Often, the protection is done through granting specific privileges to each user to create, read, update, and delete specific information.

Data Table
A Siebel table with a Type property value of Data (Public) or Data (Private). Data tables are in contrast to extension and interface tables, among others. See also interface table and table.

DBA
See DATABASE ADMINISTRATOR.

DBMS
See DATABASE MANAGEMENT SYSTEM.

Debugger
See SIEBEL VB DEBUGGER.

Dedicated Client
A computer that is directly connected to a high-speed, high-bandwidth network, able to communicate in real time with its servers. Typically, a stationary client is a desktop workstation located close to its server and benefits from rapid response times to service requests. See also MOBILE CLIENT.

Default
An operation or value that the system or application assumes, unless a user makes an alternative choice. The default is built into the program when a value or option must be assumed for the program to function.

Destination Column
The column in the destination table in a join specification. The join operation is performed on this column. It is pointed to by the foreign key (source field) in the source table. See also FOREIGN KEY COLUMN.

Destination Table
The table in a join that is joined to a business component's base table. See also BUSINESS COMPONENT OBJECT TYPE and JOIN OBJECT TYPE.

Detail
Pertains to table rows or business component records. In a one-to-many relationship, the detail records or rows correspond to the "many." For example, in a master-detail view displaying an Account form applet at the top and a Contact list applet at the bottom, the records in the Contact list applet are the detail records. See also MASTER.

Detail View Option
In Siebel Tools, one of three ways in which the object types can be displayed in the Object Explorer. When the Detail view option is selected, object definition names are displayed in the tree structure beneath their object types. See also OBJECT EXPLORER.

Details Visualization View
One of three visualization views in Siebel Tools. It is a diagram that displays how business components map to other business components and underlying tables. See also FIELD OBJECT TYPE, HIERARCHY VISUALIZATION VIEW, and RELATIONSHIP VISUALIZATION VIEW.

Dialed Number Identification Service
Technology that passes the phone number a caller dialed along with the call. In a call center handling multiple numbers such as 800 numbers, each of which may serve a different company, product line, or campaign, the DNIS number associated with a call can help determine the purpose of the call, and can be used to route the call.

Dialog Box
A box that appears on a display screen to present information or request input. Typically, dialog boxes are temporary—they disappear once the user has entered the requested information.

Directory
A way of organizing and grouping files into a hierarchical structure. The topmost directory is the root directory; the directories within a directory are subdirectories. See also SUBDIRECTORY.

Display Value
The value that is displayed to the end user. This value is language dependent. See also VALUE.

Distributed Database

A database that consists of two or more data files located at different sites on a computer network. The distributed database allows different users to access it without interfering with one another. Periodically the DBMS must synchronize all the databases to make sure that they all have consistent data.

DNIS
See DIALED NUMBER IDENTIFICATION SERVICE.

Domain
A group of computers and devices on a network that are administered as a unit with common rules and procedures. Within the Internet, domains are defined by the IP address. All devices sharing a common part of the IP address are said to be in the same domain. See also INTERNET PROTOCOL ADDRESS.

Domain Name
A name that identifies an IP address. For example, the domain name siebel.com represents the IP address 209.37.100.19. Domain names are used in URLs to identify particular Web pages. For example, in the URL http://www.siebel.com, the domain name is siebel.com. See also Internet protocol address. Domain Name System Every Web server requires a Domain Name System (DNS) server to translate domain names into IP addresses. A DNS is necessary because the Internet is based on IP addresses, not domain names. See also INTERNET PROTOCOL ADDRESS.

Drilldown (n), Drill Down (v)
A term that describes the movement from one view to another view that has greater detail.

Duplicate Conflict
A conflict that occurs when two database rows residing on different machines have the same value for the user's primary key.

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Edit Toolbar
A floating toolbar in the Siebel VB Editor that provides options for manipulating the text in the Editor window, such as cut, copy, paste, and save.

EIM
See ENTERPRISE INTEGRATION MANAGER.

Ellipsis
Symbol consisting of three dots in a row (…) seen on a button or menu option. When you click an ellipsis button, a dialog box appears.

Enterprise Integration Manager
The EIM manages the exchange of data between Siebel database tables and other corporate databases. It is used to perform bulk imports, exports, merges, and deletes. See also INTERFACE TABLE.

Escalation Policy
A set of actions that are performed when a specific set of conditions is met.

Event
1. An action or occurrence t o which a program might respond, such as key presses, button clicks, or mouse movements. 2. Siebel events refer to selected events within Siebel Applications that allow the association of extension routines that extend the base behavior of the application. When a Siebel event is "fired" or activated, the developer-specified procedures are invoked along with the standard procedures coded in Siebel Applications. 3. In Siebel CTI, an event is an occurrence, such as a call ringing, that is sent from the telephone switch to the CTI middleware, and possibly on to a Siebel client.

Extensible Markup Language
See XML.

Extension
A single line or directory number available for a teleset. The number or types of extensions available per teleset depend on the manufacturer.

Extension Column
A column that is not used by standard Siebel Applications, but is used by a custom-configured application. There are three kinds of extension columns: standard extension columns

(included for use by developers), custom extension columns in an extension table (columns added by the developer to an extension table), and custom extension columns in a base table (added by the developer to the base table of an extension table). See also BASE TABLE and EXTENSION TABLE.

Extension Table
A table in Siebel Applications that provides additional columns to a data table that cannot be directly added to the original table. The database product may only support a limited number of columns, or it may not allow adding a column to a table once it is populated with data. Siebel Applications provide extension tables for most of its base tables. See also BASE TABLE and EXTENSION COLUMN.

External Content
Information and news that is created outside of the user's company and is brought to the user by Siebel eBriefings through arrangements with external content providers (independent information providers).

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Field
1. A location in a record in which a particular type of data is stored. 2. Informally, an area for data display and entry in a Siebel applet, used synonymously with control or list column. See also FIELD OBJECT TYPE.

Field, Calculated
1. A calculated field is one whose value is derived from some formula involving other fields. You do not enter data into a calculated field; the system automatically determines the correct value. 2. A calculated field in Siebel Tools is implemented by the specification of a Calculated Value property value in a Field object definition. See also FIELD OBJECT TYPE and PROPERTY.

Field Object Type
A Siebel object type that maps a column to a business component. A field provides a meaningful name that does not require knowledge of the base table or joined table of its origin. Fields supply data to controls and list columns in the user interface. See also COLUMN OBJECT TYPE, CONTROL OBJECT TYPE and FIELD, CALCULATED.

File
A collection of data or information that has a name, called the filename. Almost all information stored in a computer must be in a file. There are many different types of files: data files, text files, program files, directory files, and so on.

File Attachment Applet
A style of applet, typically a list applet. It provides access to external documents from within Siebel applications and allows the user to open, add, edit, or remove documents from the list applet. It also provides synchronization and shared access support for attached documents. See also APPLET OBJECT TYPE.

Firewall
A system designed to prevent unauthorized access to or from a private network. Firewalls can be implemented in both hardware and software, or a combination of both.

Flat-File Database
A simple database system in which each database is contained in a single table. Relational database systems can use multiple tables to store information, and each table can have a different record format.

Flat View Option
A viewing option in the Object Explorer that displays all object types without their hierarchical relationships. Invoked by clicking the Flat tab.

Foreign Key Column
A column in a database table that is used to store the values of a primary key column of another table.

Form Applet
A style of applet in Siebel applications that displays data in a data entry form. It displays one record at a time from the underlying business component. Fields in the business component appear on the form applet as text boxes, check boxes, and other controls.

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Graphical User Interface
A GUI is a type of environment that represents programs, files, and options by means of icons, menus, and dialog boxes on the screen. Some of the GUI items found in Siebel applications include the view bar, toolbar, and tab bar, and various dialog boxes, controls, and buttons.

Group Control
A rectangular box that is drawn on the applet and serves as a visual aid for organizing the controls in the applet. It does not have data-editing or display capabilities.

GUI
Pronounced goo-ee. See GRAPHICAL USER INTERFACE.

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HelpId
A value specifying a topic name for context-sensitive Help. To make context-sensitive Help work for an applet, the applet needs to pass the HelpId value to Windows Online Help.

Hierarchy Visualization View
In Siebel Tools, one of three visualization views. It is a diagram that displays the parent-child relationships between the selected object definition, and its parent and child object definitions, and their parent and child object definitions (and so on). See also DETAILS VISUALIZATION VIEW and RELATIONSHIPS VISUALIZATION VIEW.

Home Page
The starting point of a Web site. Also, the Web page that a browser is set to use when the user first starts it.

HTML
See HYPERTEXT MARKUP LANGUAGE.

HTML Application
A Siebel Repository representation of a Siebel HTML Thin Client-based Application as an object of type Application. It contains definitions of the various Applets, Views, Web pages, and so forth that make up the Application.

HTML Template
An HTML model or style file for generating a Web Template containing both standard HTML and Siebel Tags. Information in the HTML Template file is combined with information stored in the Web Template Object definition in the repository to generate the final HTML output.

HTTP
See HYPERTEXT TRANSFER PROTOCOL.

HTTPS
See SECURE HYPERTEXT TRANSFER PROTOCOL (S-HTTP).

Hypertext
A method of presenting textual information online so that users can follow built-in links (provided by the author) to take them from topic to topic. Rather than being restricted to viewing information in a certain sequence, users are free to choose which links they want to follow in order to browse through related topics in the order they choose. Siebel supports hypertext links to supplement navigational techniques and integration with attachments.

Hypertext Markup Language
HTML is the coding language used to create hypertext documents for use on the World Wide Web. In HTML documents, blocks of text are surrounded by codes (called tags) that indicate how the text should be displayed and where links are placed. HTML is platform-independent, meaning that the codes can be read by Web-browser programs on any operating system.

Hypertext Transfer Protocol
HTTP is the communications standard used to connect World Wide Web sites and to move hypertext files across the Internet.

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Identifier Code
A unique code in Siebel InterActive used by a content provider to label a company about which it provides news and other information. Each external content provider uses a different coding scheme to label the companies it tracks.

Image Control
An image control in an applet lets the developer display pictures stored as bitmap files, which are associated with business component records.

Import
To use data produced by another application.

Import Aux Field Object Type
A Siebel object type that defines the auxiliary key field name for the business component specified in an Import Object object definition. It is a child of the object type called Import Object.

Index
In relational databases, a system table that cross-references the values in selected database columns to the rows in which those values reside. Indexes significantly improve the performance of queries.

Instantiate
1. In object-oriented programming, instantiate refers to creating an instance of a class. 2. In the Siebel Object Interfaces, instantiate refers to creating an instance of an object definition.

Interactive Voice Response
Telephony technology that allows a customer to interact with the switch by using touch-tone or voice commands to navigate through a menu hierarchy or enter customer-specific information. Also referred to as an IVRU (Interactive Voice Response Unit).

Interapplication Variables
Variables that may be sent back and forth between a Siebel application and applications that are accessing the Siebel application. interface table A table that denormalizes a group of base tables into a single table that external programs can interface to. Used by the Siebel Enterprise Integration Manager to import initial data for populating one or more base tables. See also base table and table.

Internet
The worldwide collection of networks and gateways that use the TCP/IP suite of protocols to communicate with one another.

Intersection Table
A table in Siebel applications that implements a many-to-many relationship between two business components. It may also hold intersection data columns, which store information pertaining to each association. Intersection tables provide the data structures for association applets. See also ASSOCIATION APPLET.

Intranet
A private network inside a company or organization that uses the same kinds of software (such as Web servers and browsers) that a user would find on the public Internet. Access to information on a corporate intranet is generally restricted to people within the organization. An organization's intranet may or may not also provide employees with access to the Internet.

Internet Protocol Address
A 32-bit binary number that uniquely identifies a host (computer) connected to the Internet or to other Internet hosts, for the purpose of communication through the transfer of packets. See also DOMAIN, DOMAIN NAME and DOMAIN NAME SYSTEM.

IVR/IVRU
See INTERACTIVE VOICE RESPONSE.

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Join
In relational databases, a query that retrieves data from two separate tables. The two tables must be joined by at least one common column. The join column is a member of both tables. See also JOIN OBJECT TYPE.

Join Object Type
A Siebel object type that creates a named association between two tables. One table is the base table for the parent business component of the join. Because of a foreign key column in the joined table, a join can provide rows on a one-to-one basis to the business component even though the two tables do not necessarily have a one-to-one relationship. Join is a child object type of the Business Component object type.

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Label Control
This control is used to place words on the applet surface. It is a visual aid and has no data display or entry capabilities.

LAN
See LOCAL AREA NETWORK.

Language Code
The code that specifies what human language is used for a display value. In the Language Administration view, each language has its own code identifier. For example, ENU means English U.S.

Library
In programming, a collection of routines stored in a file. Each set of instructions in a library has a name, and each performs a different task.

Link Object Type
A Siebel object type that implements a one-to-many relationship between business components based on their base tables. The master and detail business components are specified in the Link object definition and the field on which the two are linked.

Linguistic Search
One of the searches performed for all forms of a word; for example, run, runs, running, ran.

List Applet
A style of applet that displays data in a scrolling list table format, similar to a spreadsheet or word processor table. Rows in the list applet correspond to records in the business component; list columns in the list applet correspond to fields in the business component.

List-Form View
In a list-form view, two applets show data from the same business component. The lower form applet displays an expanded list of information for the contents of the selected record in the upper list applet. See also MASTER-DETAIL VIEW.

List of Values
The LOV is a set of values that populates a static picklist, configured in the Application Administration view of Siebel applications.

Literature Types
Categories for the organization of literature stored in the Siebel Encyclopedia. Examples of literature types include Presentations, Training, and Recommended Links to Web sites of interest.

Local Area Network

A computer network that spans a relatively small area. Most LANs are confined to a single building or group of buildings.

Local Database
A database for a mobile client, which will periodically be synchronized with a corporate database.

Local Lock
In Siebel Tools, the lock placed on a project in the local repository, instead of the server. Using a local lock enables developers to make modifications to object definitions in the project on the local database. It is advantageous for testing, but it does not allow the developer to check in the modified object definitions to the server. See also CHECK IN (v), CHECK-IN (n).

Locking
In Siebel Tools, locking is similar to checking out. When a project is locked, one developer can make modifications to its object definitions, but others cannot. However, unlike check-out, locking a project does not copy its object definitions from the server to the local database. See also CHECK OUT (v), CHECK-OUT (n).

Logical Operators
See BOOLEAN OPERATORS.

Logical Schema
Provides a model of the total database and acts as an intermediary between the two other types of schemas (internal and external) that deal with storing information and presenting it to the user. See also SCHEMA.

LOV
See LIST OF VALUES.

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Manager Visibility
A setting that specifies a manager's access to records created by, or assigned to, the sales teams that report to that manager.

MAPI
See MESSAGING APPLICATION PROGRAMMING INTERFACE.

Marketing File Server
A file server that contains the marketing literature and correspondence files for client access. Also known as the Siebel File Server.

Master
Term that pertains to table rows or business component records. In a one-to-many relationship, the master records or rows correspond to the "one." For example, in a master-detail view displaying an Account form applet in the upper portion of the view and a Contacts list applet in the lower portion of the view, the record in the Account form applet is the master record.

Master-Detail View
This style of view has two applets displaying data from a pair of business components in a one-to-many relationship (implemented with a link). The upper form applet displays data from one record in the master business component, and the lower list applet displays a list of records from the detail business component. See also list-form view.

Menu Bar
A horizontal bar at the top of a window, which contains the names of the menus available in the application.

Menu Designer
A design tool in Siebel Tools for specifying the menus and submenus for an application or applet. It is invoked from the shortcut menu that is displayed by right-clicking an Applet or Application object definition in the Object List Editor.

Messaging Application Programming Interface
MAPI is a system built into Microsoft Windows that enables different email applications to work together to distribute mail. As long as both applications are MAPI-enabled, they can share messages with each other.

Method

A software routine that is invoked in response to an event. The standard methods in Siebel Tools provide a variety of common functions, such as adding or deleting a record in response to a button click. Customized methods are programmed using the Siebel VB Editor.

Mobile Client
Typically, refers to a portable computer used by a field representative who does not have continuous access to a network. The client (field representative) must periodically access the Siebel Remote Server via a dialup, WAN, or LAN connection to synchronize data changes with the Siebel database (on the database server) and Siebel File System. See also DEDICATED CLIENT and THIN CLIENT.

Model
A representation of products, services, and the relationships among them that is used to configure a customer solution. Models are designed in the Configuration Designer screen and are used in the Quote Assistant. The choices and constraints embodied in the model guide Quote Assistant users toward viable solutions. Quote Assistant users choose the model they want to use to configure a solution. See also RELEASE and RULE.

Multi-Value Group
The MVG is a set of Siebel detail records attached to the current record in the business component that holds the corresponding multi-value field. The MVG records are displayed and maintained by means of a MVG applet.

Multi-Value Group Applet
A style of applet that displays and enables the maintenance of an MVG. It is typically implemented as a dialog box containing a scrolling list table, such as appears in a list applet. For example, the ellipsis button adjacent to an Address text box control invokes an MVG applet that displays the street address, city, state, and zip code for all of the addresses of the account.

MVG
See MULTI-VALUE GROUP.

MVG Applet
See MULTI-VALUE GROUP APPLET.

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Name
A sequence of one or more characters that uniquely identifies a file, variable, account, or other entity. Computer systems impose various rules about naming objects. For example, there is often a limit to the number of characters you can use, and not all characters are allowed.

Needs Analysis
A needs analysis enables agents to ask questions and assign scores to the answers that customers give them. Using this information, agents can recommend a package of products or a single product and cross-sell groups of products or a single product.

Network
A group of two or more computers connected so that they can share resources.

New Applet Wizard
An interactive utility within Siebel Tools that guides the developer through each step of creating a list, form, chart, or tree applet.

News Archive
An archive of news articles for the last 90 days about the user’s company, accounts, and competitors that is found in Siebel InterActive.

Non-Primary (NP) Forecasting
Allows any opportunity or product sales team member to enter a forecast. Use the Opportunity Forecasts (NP) and Product Forecasts (NP) views to enter non-primary forecasts.

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Object
A self-contained entity that consists of both data and procedures to manipulate data. Found in object-oriented programming.

Object Compiler
Allows the developer to create an .srf file from the current set of object definitions in the local repository. This file can be executed by Siebel applications software at a client computer. See also .SRF FILE.

Object Definition
A construct that specifies one user interface, abstract data representation, or direct database representation piece of the software. It is a named set of properties that belong to a specific object type, and it resides in a Siebel repository. An object definition's object type specifies the set of properties requiring values. Object definitions are listed in the Object List Editor for the object type currently selected in the Object Explorer. For example, the object type Business Component has the object definitions Account, Contact, Opportunity, and many others.

Object Explorer
One of three windows (the other two are the Object List Editor and the Properties window) that collectively provide the bulk of the functionality of Siebel Tools. The Object Explorer displays the object types in the repository in a hierarchical tree structure, similar in appearance to the Windows Explorer. Each folder icon corresponds to one object type. See also OBJECT LIST EDITOR and PROPERTIES WINDOW.

Object Interface Event
One of various events within Siebel Applications that allow the association of extension routines (event methods).

Object List Editor
One of three windows (the other two are the Object Explorer and the Properties window) that collectively provide the most of the functionality of Siebel Tools. The Object List Editor lists the object definitions for the currently selected object type in the Object Explorer. Each column in the scrolling list corresponds to a property; each row corresponds to an object definition. See also OBJECT EXPLORER and PROPERTIES WINDOW.

Object Model
See SIEBEL OBJECT MODEL.

Object Style
An informal descriptive category of object definitions within an object type. The name of the object style is followed by the name of the object type. For example, list applet, where list is the object style and applet is the object type.

Object Type
A named structure from which object definitions of the same type can be created. An object type has a predefined set of properties. The object definitions created from it have values for each of these properties (default or user-specified). The properties correspond to the columns in the Object List Editor. Object types can have parent and child object types that form a hierarchical tree structure, which is visible in the Object Explorer. Specific object types are described in this Glossary under their object type names; for example, List object type.

OLAP
See ONLINE ANALYTICAL PROCESSING.

OLE Application Server
A mode of Siebel object interface operation. It allows an external program to access the objects in Siebel Applications while the user interface is enabled. It also allows Siebel Applications to run normally, while providing access by external applications to the objects.

OLE Data Server
A mode of Siebel object interface operation. It allows an external program to access the objects in Siebel Applications while the user interface is disabled. In this mode, data manipulation operations can take place, but user interface operations cannot.

One-to-Many Extension Table
An extension table whose rows are the detail rows of a one-to-many relationship with master rows in a base table. See also EXTENSION TABLE.

One-to-Many Relationship
A relationship between two kinds of entities where one of the first kind relates to one or more of the second kind. For example, a purchase order has one or more line items. Entity relationship diagrams use symbols to indicate one-to-one and one-to-many relationships between database tables or logical entities.

One-to-One Relationship
A relationship between two kinds of entities where one of the first kind corresponds to one of the second kind. For example, an account has one account manager.

Online Analytical Processing
OLAP is a category of software tools that provides analysis of data stored in a database. OLAP tools enable users to analyze different dimensions of multidimensional data. For example, it provides time series and trend analysis views. The OLAP server understands how data is organized in the database and has special functions for analyzing the data.

Operation Conflict
A conflict where one database transaction attempts to update a row while another attempts to delete it.

Operators
Symbols or other characters indicating an operation that acts on one or more elements in mathematics, programming, and computer applications. The symbols may include the standard mathematical symbols (+, -, /, ×), Boolean operators such as AND, OR, and NOT, and relational operators such as >, <, and /. See also BOOLEAN OPERATORS and RELATIONAL OPERATORS.

Operator	Mnemonic	Meaning
=	EQ	Equal to
<> (or !=)	NE	Not equal to
>	GT	Greater than
>=	GE	Greater than or equal to
<	LT	Less than
<=	LE	Less than or equal to

Opportunity
A potential revenue-producing event. originating applet The applet containing the control or list column that invokes a list, MVG (multi-value group), or detail applet.

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Package
A set of compressed files that contain transactions to be uploaded during database synchronization.

PAP
See PASSWORD AUTHENTICATION PROTOCOL.

Parameter
1. In programming terminology, a value that is given to a variable, either at the beginning of an operation or before an expression is evaluated by a program. Often used synonymously with argument. 2. A characteristic or property of an entity such as, for example, a Siebel Server component, or a CTI configuration. Configuring Siebel software involves specifying values or strings for certain named parameters. See also ARGUMENT and VALUE.

Parent
The hierarchically higher member of a special relationship between two object types or object definitions called the parent-child relationship. See also child, parent object definition, and parent object type.

Parent-Child Relationship
The hierarchical relationship between two object types or object definitions in which there is a one-to-many relationship between the parent and child. Each parent-child relationship between two object types maps to a one-to-many relationship between each object definition of the parent type and multiple object definitions of the child type.

Parent Object Definition
The hierarchically higher member of a parent-child relationship between object definitions. A parent object definition has multiple child object definitions (within one parent-child relationship between a pair of object types). For example, the Opportunity Form Applet object definition has 39 child object definitions of the Control object type, including Account, CloseDate, and Description.

Parent Object Type

The hierarchically higher member of a parent-child relationship between object types. A parent object type has one or more child object types. For example, the Applet object type has nine child object types, including Applet Script, Control, Chart, List, and Tree.

Password Authentication Protocol
PAP is the most basic form of authentication, in which a username and password are transmitted over a network and compared to a table of encrypted name-password pairs.

Path
A sequence of symbols and names that identifies a file. Every file has a name, referred to as a filename, so the simplest type of pathname is just a filename.

Personal Visibility
A term that refers to the user's access to those records that were created by, or assigned to, the user.

Physical Schema
A view of the physical files that make up a database. The schema includes filenames, file locations, access methodology, and actual or potential data derivations. See also LOGICAL SCHEMA, SCHEMA.

Pick Applet
A style of applet that resides in a dialog box. The dialog box appears when a user must make a selection in a control or list column that is designated to activate that pick applet. The pick applet provides a list or table of
selection values from which a user selects a value or row.

Picklist
A user interface feature that displays a list of valid values for a specified field or list column cell.

Plug-In
An application that plugs in to another primary application to extend its capabilities. Plug-ins are often developed by third-party manufacturers according to technical specifications provided by the primary application's manufacturer.

Portfolio
A portfolio in Siebel Finance is a collection of bank accounts, which can be either loans or deposits. Within the context of Siebel Finance, portfolio does not refer to an investor's stock and bond holdings.

Postoperation Event
An event that occurs after a standard operation in Siebel Applications. Useful when the event relies on data that may have been updated in the course of executing the standard event in the application, or when standard processing is desired no matter what the outcome of the custom procedure. See also PREOPERATION EVENT.

Predictive Dialer
Telephony technology that reads contact information from a database and initiates outbound calls. The system can detect differences between busy signals, out-of-service numbers, answering machines, and actual human responses, which are passed to a call center agent.

Preoperation Event
An event that occurs before a standard operation in Siebel Applications. For example, the PreDeleteRecord event occurs before the DeleteRecord event. See also POSTOPERATION EVENT.

Primary Forecasting
Allows only the primary representative for each opportunity or product sales team to enter a forecast.

Primary Key Column
A column (or member of a group of columns) that uniquely identifies each row, and has been chosen as the key for the table. Also called primary key.

Product
An individual part that is included in the model and appears in Siebel Quote Assistant. Products can be physical devices or intangible products such as service agreements, sales promotions, or bundles. The same product can appear in several models or more than once in a single model. Products must be entered into the Siebel database before they can be added to a model. See also DATABASE INITIALIZATION, MODEL and SIEBEL DATA MODEL.

Product Configurator
See SIEBEL PRODUCT CONFIGURATOR.

Project
In Siebel Tools, a named set of object definitions that can be checked out for modification, at one time, by a developer. See also CHECK OUT (v), CHECK-OUT (n).

Properties Window
One of three windows that collectively provide the core functionality of Siebel Tools. (The other two windows are the Object Explorer and the Object List Editor.) The Properties window lists the property values for the currently selected object definition in the Object List Editor. See also OBJECT EXPLORER and OBJECT LIST EDITOR.

Property
1. In object-oriented programming, a named characteristic of an object. 2. In Siebel Tools, a named characteristic of the software construct that an object definition defines. The properties of object definitions correspond to columns in the Object List Editor, and to the list of property values appearing in the Properties window for the currently selected object definition. 3. In Siebel Product Configurator, a configuration-related characteristic of a category or product. The value of the property can be a number or a text string.

Protocol
1. Technical standards designed to enable computers to connect with each other and to exchange information. 2. An agreed-upon format for transmitting data between two devices. The protocol determines the type of error checking to be used, the data compression method, if any, how the sending device will indicate that it has finished sending a message, and how the receiving device will indicate that it has received a message.

Publish
1. In Siebel eBriefings, publish means to make information available to end users in Briefings. 2. In Siebel Remote, publish refers to the distribution of files from the Siebel File System to the local file system in order to have the files available for the mobile user during the next synchronization session.

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QBE
See QUERY BY EXAMPLE.

Query
1. In Siebel applications, a request for information from a database. Siebel uses the following methods for posing queries: query by choosing parameters, query by example (QBE), and query by language. 2. An operation performed on one or more tables. A query returns a temporary table called a results table or dataset, or performs updates on the original table or tables.

Query by Example
1. QBE is a method of forming queries by entering conditions in the blank fields of records displayed by the database program. 2. In Siebel applications, QBE is the method employed for querying the records displayed in a list or form applet. 3. In Siebel Tools, QBE is the method employed for creating Object List Editor queries.

Query Language
A specialized language for requesting information from a database. One of the standard query languages is Structured Query Language (SQL).

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Radio Buttons
A button in a graphical user interface that appears as a small circle next to the name of the button. Only one button in a set of radio buttons can be selected at a time; so when the user selects one button, the other buttons in the set are automatically deselected.

RDBMS
Relational Database Management System. (RDBMS is pronounced as separate letters.) See RELATIONAL DATABASE and RELATIONAL DATABASE MANAGEMENT SYSTEM.

Record
1. In second- and third-generation programming terminology, a record is a set of field values corresponding to a single entity, such as a customer account or employee, in a sequential or indexed file. 2. A set of values in Siebel applications for a particular business component. These values are displayed as a row in a list applet, or as a set of data entry controls in a form

applet.3. Sometimes used as a synonym for row. See ROW.

Record Access Visibility
A set of restrictions on a user's access to data records in the server database. Access is based on the user's ownership of records resulting from record creation or assignment, membership on sales teams or contact access lists, and position in the organization.

Regional Database
A Siebel database, typically at a remote site, to which data is replicated from a parent database by Siebel Replication Manager. The parent database may be a master database located at a headquarters site or may be another regional database. A regional database is automatically synchronized with its parent database via the Replication Manager, and optionally may be a full replica of its parent database. See also MOBILE CLIENT.

Relational Database
A database system that stores data in the form of tables, and conducts searches by using data in specified columns of one table to find additional data in another table. The rows of a table represent records and the columns represent fields. In conducting searches, a relational database matches information in a corresponding field of another table to produce a third table that combines requested data from both tables. See also RELATIONAL DATABASE MANAGEMENT SYSTEM.

Relational Database Management System
A database management software product that implements the relational model, which is a logical data model in which all data is represented as a collection of related tables. See also RELATIONAL DATABASE.

Relational Operators
Operators that compare two values. Relational operators are sometimes called comparison operators. Expressions that contain relational operators are called relational expressions. See also BOOLEAM OPERATORS and OPERATORS.

Relationship Visualization View
Roles are a pre-configured collection of permissions that are normally linked to users and groups. Content DB has a set of standard roles and hidden roles that are provided out of the box however an unlimited number of roles can be One of three visualization views in Siebel Tools. It is a diagram that displays how a selected business component links to other business components by means of a multi-value link object definitions, or how a selected table joins to other tables by means of Join object definitions. See also DETAILS VISUALIATION VIEW and HIERARCHY VISUALIZATION VIEW.

Release
The process by which model data is made available to Siebel Quote Assistant users.

Report
A hard copy or electronic representation of the information displayed by a Siebel application.

Repository
The complete set of object definitions defining a Siebel application or suite of applications (such as the Siebel Service or Siebel Applications). It is part of the Siebel client, compiled into the Siebel repository file (the .srf file). The master copy of the repository resides on a server database where multiple developers can access it to make changes and additions. A local copy resides on a developer's client machine for development and testing.

Repository Migration
The process of relocating a customized repository from one database to another, such as from the development database to a test environment or, subsequently, from development to production.

Repository Search
A facility in Siebel Tools that provides the means to search for object definitions in the repository. The search is based on text in the name of the object or on other properties.

Resource
A value or total that is defined in the Configuration Designer screen and monitored in Siebel Quote Assistant. Resource totals change dynamically when the user picks items that have Provides and Consumes rules.

Responsibility
A class of users requiring access to the same set of application features. A user's responsibility or responsibilities identify the views that he or she can navigate to in Siebel Applications.

Routing
A category of restrictions on a remote user's access to records. This restriction is in addition to record access visibility. See also manager visibility, record access visibility, and sales team visibility.

Routing Rules
Rules used by Siebel Remote and Replication Manager to determine what data and transactions are routed to mobile client databases and regional databases. See also REGIONAL DATABASE.

Row
A row in a database table. It holds data about a specific business entity of the type stored in the table, such as a single purchase order, employee, or inventory item found in the purchase orders, employees, or inventory tables. Rows in tables are comparable to records in indexed files. See also RECORD and TABLE.

Rule
A way to define the relationships among products and categories within the model. There are four rule types: Requires, Excludes, Provides, and Consumes.

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Sales Team
A list of user positions in an account, opportunity, or contact record that provides sales team visibility to the record for the listed users. Sales teams are updated through Assignment Manager. In a contact record only, a sales team is called a contact access list.

Sales Team Visibility
A user's access rights to records based on his or her presence on sales teams and contact access lists.

Schema
A description of a database to a database management system (DBMS) in the language provided by the DBMS. A schema defines aspects of the database, such as attributes (fields), domains, and parameters of the attributes. See also DATABASE MANAGEMENT SYSTEM and LOGICAL SCHEMA.

Screen
1. The display part of a monitor. 2. In Siebel applications, a screen is a logical collection of views. A list of the views available for the screen appears when you click a tab in the tab bar or choose a menu option in the Screens menu.

Screen Pop
The process of a screen appearing on an agent's machine, generally about an incoming call or caller.

Screens Menu
The drop-down menu that appears when you click Screens on the menu bar. It lists the available screens in the application. When you choose a screen from the menu, the list of views available for that screen appears in a submenu.

Script
1. A procedure written in Siebel VB and attached to an object definition, such as an Application, Business Component, Applet or Control Object Definition. 2. Siebel's SmartScript and Script Wizard in Siebel Service or Siebel Call Center. See SIEBEL SMARTSCRIPT.

Scripted Flag
A property called Scripted in the Object List Editor. It appears for object definitions that can have a Siebel VB script attached to them and indicates whether one or more Siebel VB scripts are attached to the object definition.

Search Specification
A conditional expression in the Search Specification property in an Applet, Business Component, Link, or Pick List object definition. It imposes restrictions on the set of records being retrieved.

Sector
A rectangular area of a view. An applet is positioned in one or more contiguous sections. A view displays either six or eight sectors, depending on the size of the monitor displaying the view.

Secure Hypertext Transfer Protocol
An extension to the HTTP protocol to support sending data and individual messages securely over the World Wide Web. Also called HTTPS.

Security
A user's ability to view portions of database for Siebel applications. Typically, security is based on a need to know, as specified by company management.

Seed Data
The supply of content from a variety of sources, usually via a paid service subscription for reuse and integration with other content. A new functionality in some ECM tools, allowing users to subscribe to folders and change events, which is Required data created in the Siebel database during a Siebel applications installation process that is needed for the application to start up.

Selective Routing
A routing capability that provides the means to restrict the list of accounts routed to individual salespersons who are connected to the server remotely. This feature makes it possible to limit the connection time and disk space consumption of mobile users. Also called SELECTIVE RETRIEVAL.

Server
1. On a local area network, a computer that controls access to all or part of the network and its resources (such as, software, hardware and data). A computer acting as a server makes resources available to computers acting as workstations on the network. 2. Also, a software program that provides a specific kind of service (resource) to client software running on other computers.

Server Lock
In Siebel Tools, a check-out option that determines whether a specified project or projects should be locked on the server as a part of the check-out process. This is the normal setting for check-out of projects. The alternative is local locking. See also LOCAL LOCK.

Server Manager
One of the primary architectural components of Siebel applications; the control and monitoring component.

Server Wins
A rule used to resolve a data conflict in which the data value from the server database takes precedence over the data value on the client database. See also CLIENT WINS.

S-HTTP
Also HTTPS. See SECURE HYPERTEXT TRANSFER PROTOCOL.

Siebel Call Center
This application enables users to provide sales, marketing, and customer service assistance to customers. It accesses the full power of Siebel Sales and Siebel Service to integrate all available customer information.

Siebel CTI
Siebel CTI integrates front office application software from Siebel Systems with your telephone switch and CTI middleware. With Siebel CTI, the telephone system provides notification of telephony events to the Siebel desktop, and the Siebel applications in turn control or interact with the phone system. See also COMPUTER TELEPHONY INTEGRATION.

Siebel Data Model
A data model that defines how the data used by Siebel Applications is stored in a standard third-party relational DBMS such as Microsoft SQL Server, Oracle, Informix, or Sybase. It also specifies the correct methodology to reconfigure the database.

Siebel Applications
An integrated suite of Siebel eBusiness, marketing, sales, contact management, and call center management applications from Oracle. Depending on context, the term Siebel Applications can refer to individual applications in the suite or to the entire Siebel application suite.

Siebel Field Service
A product that provides complete field service functionality for call center agents and mobile field service engineers. It builds on Siebel Service to handle a full complement of field service scenarios and details, including the required skills, tools, and parts for all service activities.

Siebel Industry Applications
A suite of industry-specific applications for the communications, consumer sector, energy, finance, governmental, high technology, insurance, and life sciences industries.

Siebel Marketing
This application enables marketing professionals, sales and service managers, and business analysts to monitor overall company performance and the effectiveness of company programs and activities. It extracts information from Siebel Sales, Siebel Service, and Siebel Call Center. The information is compiled into a customer data mart designed for fast data analysis. See also DATA MART.

Siebel Object Interfaces

An application program interface that supports the Component Object Model (COM). The developer uses it to access Siebel objects by capturing Siebel events and invoking Siebel object methods from within other applications and programming languages that can host OLE automation servers.

Siebel Object Model
A structure and methodology that facilitates the creation of complete applications through the creation and modification of object definitions.

Siebel OLE Application Server
See OLE APPLICATION SERVER.

Siebel OLE Data Server
See OLE DATA SERVER.

Siebel Product Configurator
A complete model design, configuration, and quotation system. It has two parts: Siebel Configuration Designer and Siebel Quote Assistant.

Siebel Proposal Generator
A product that enables sales professionals to build sales proposals tailored to customer requirements and based on customer profile.

Siebel Quote Assistant
A feature used to configure solutions and prepare quotes that are ready for distribution to the customer. It is part of Siebel Product Configurator.

Siebel Remote
A product that enables mobile computing by providing a complete exchange of information between a mobile computer and the corporate server. Mobile users can access Siebel applications through the server.

Siebel Repository File
An .srf file is a compiled repository file. When it is opened by the Siebel application engine, the Siebel repository file provides all of the object definition information required for interaction with the enterprise data and users. See also REPOSITORY.

Siebel Sales
This application enables teams of sales and marketing professionals to manage sales information throughout the entire sales cycle.

Siebel Server
This server supports the processes for Siebel clients. Siebel Servers Provides both a command-line interface and a graphical user interface to provide full remote management of all Siebel servers across the enterprise. Siebel administrators, or external applications using the command-line interface can administer and access comprehensive performance statistics for all server processes through the Siebel Server Manager.

Siebel Service
This application enables teams of customer service, sales, and marketing professionals to ensure complete customer satisfaction by using closedloop, service request, management capabilities.

Siebel SmartScript
A Siebel option that enables business analysts, call center managers, and Siebel developers to define the application workflow for an interactive situation in a script. These situations include both inbound communications and outbound contacts. The workflow of the interaction is controlled by the script, not by the agent.

Siebel Tag
An HTML tag (part of a library of Siebel-developed tags) that can be embedded in a standard HTML file to give directives to the Siebel Web Engine for creating the HTML pages that are displayed to users. An example is a Placeholder for a Siebel Control or Applet.

Siebel Tools
A Siebel software configuration toolset used by qualified developers to perform declarative software development; that is, the creation and modification of object definitions and their properties. It enables developers to reconfigure and extend Siebel Applications.

Siebel VB
The programming language in Siebel Tools that is used to write custom procedures to attach to certain object types. It includes a code editor, debugger, and interpreter-compiler. See also SIEBEL VB COMPILER, SIEBEL VB DEBUGGER and SIEBEL VB EDITOR.

Siebel VB Compiler

A nonvisual component of Siebel VB that compiles and executes Siebel VB custom program routines. It is similar to Microsoft's Visual Basic Language Interpreter. See also SIEBEL VB, SIEBEL VB DEBUGGER and SIEBEL VB EDITOR.

Siebel VB Debugger
A facility in Siebel Tools for locating and diagnosing errors in Siebel VB scripts. It uses the Siebel VB window, plus a diagnostic window to display program variables and their values. See also SIEBEL VB, SIEBEL VB COMPILER and SIEBEL VB EDITOR.

Siebel VB Editor
An integrated code editor used to create, view, edit, and save custom program routines written in Siebel VB. See also SIEBEL VB, SIEBEL VB COMPILER and SIEBEL VB DEBUGGER.

Siebel Workflow Manager
A module that can be used with any Siebel application to enable automated business policies and procedures for improved customer service sales processes.

Solution
A configured set of products and services based on a single model. Solutions are produced in the Quote Assistant, which is part of Siebel Product Configurator. The Quote Assistant user decides which model to use for each solution. See also MODEL.

SQL
See STRUCTURED QUERY LANGUAGE.

.srf File
See SIEBEL REPOSITORY FILE.

SSL
See SECURE HYPERTEXT TRANSFER PROTOCOL.

Standard Table
A table included in standard Siebel Applications database.

State Machine
An engine that enforces the transitions between states for an object during its lifetime. A state represents the status of an object, such as open, closed, pending, and where the object is at each point in its lifetime. The state can also control whether or not the data associated with objects can be modified.

State Model
A description of acceptable states and state transitions that the state machine enforces. The state machine ensures that objects (such as Service Request or Product Defects objects) go through the desired process defined in the state model. After a state model is created, it is immediately activated; no additional steps are required.

State Transition
Definition of the allowable migration of an object from one state to the next.

State Picklist
A selection list invoked when you click the ellipsis button in some text boxes and list columns. This picklist is an alternative to a pick applet. It is used when the selection list is a static list of values rather than one generated dynamically from the contents of business component records.

Structured Query Language
SQL is an industry-standard language for constructing queries for database tables in an RDBMS.

Style
See OBJECT STYLE.

Subdirectory
A directory located beneath another directory. Every directory, except the root directory, is a subdirectory.

System Administrator
A person responsible for the entire Siebel system, including installing, maintaining, and upgrading Siebel products as needed. See also APPLICATION ADMINISTRATOR and DATABASE ADMINISTRATOR.

System Primary Key

In a database table, a column whose value is a unique identifier for the row. This is a system-generated and maintained value. See also USER PRIMARY KEY.

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Tab Bar
In a database table, a column whose value is a unique identifier for the row. This is a system-generated and maintained value. See also user primary key.The horizontal row of tabs just beneath the menu bar in a Siebel application window. Each tab corresponds to a screen whose default view is displayed (activated) when the tab is clicked.

Table
A data structure in relational databases, which is characterized by rows and columns with data occupying, or potentially occupying, each cell formed by a row-column intersection. See also TABLE OBJECT TYPE.

Table Object Type
A Siebel object type that maps a physical table in the DBMS, and it has Column and Index Column child object definitions that represent the database tables, columns, and indexes. See also ASSOCIATION TABLE, BASE TABLE, DATA TABLE, DESTINATION TABLE, EXTENSION TABLE, INTERSECTION TABLE, ONE-TO-MANY EXTENSION TABLE and STANDARD TABLE.

Teleset
Telephone supporting multiple extensions and controls that issue commands to the switch, such as login, make call, answer call, hold call, conference, and transfer. The Siebel CTI toolbar provides access to many of these commands.

Text Box Control
A type of control that displays text and permits data entry.

Thin Client
A deployment option that enables the user interface (the client) of Siebel applications to appear in a Web browser, while most of the application logic resides on the server. Siebel offers three types of thin clients—Siebel Thin Client for Windows, Siebel HTML Thin Client, and Siebel Java Thin Client.

Tracing
A debugging feature in the Siebel Object Interfaces. It is activated by the TraceOn method and deactivated by the TraceOff method. While tracing is active, the allocations and deallocations of Siebel objects are recorded in a trace file.

Transaction
A discrete activity within a computer system such as an entry of a customer order or an update of an inventory item.

Transaction Log
A file that records transactional changes occurring in a database. It provides a basis for updating a master file and establishing an audit trail.

Transaction Processing
A processing method in which transactions are executed immediately after they are received by the system.

Tree
A data structure containing zero or more nodes linked together in a hierarchical fashion.

Tree Applet
A style of applet used to create a view (called an Explorer view). An Explorer view allows the user to navigate hierarchically through a structured list of business component records from one or more business components.

Types View Option
This option is found in the Object Explorer and is one of three ways to display object types. When the Types view option is active, object type names appear hierarchically in the tree structure.

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Uniform Resource Locator
A URL is an address on the World Wide Web.

Unvalidated Resource
A resource whose value can become negative in Siebel Quote Assistant. See also validated resource.

Update Conflict
A conflict where two database change transactions attempt to update the same field in the same row. See also conflict detection and conflict resolution.

Upload
The process of transmitting information from the client to the server.

URL
See UNIFORM RESOURCE LOCATOR.

User Interface Object Definitions
Object definitions that reside in the User Interface Objects layer (in Siebel Tools). They define the visual interface that the user sees and interacts with. Data from business object definitions is presented to the user for viewing and modification by means of user interface object definitions. These include the Applet object type and its child object types, View and Screen, and the object types used to implement toolbars, menus, and the Find dialog box.

User Interface Objects Layer
The architectural layer in the Siebel repository that includes user interface object definitions. These object definitions are used to present and manipulate data from object definitions in the Business Objects Layer. Object types in this layer include Application, Applet, Control, Screen, View, List Column, and so on. See also object definition.

Username
A name used to log on to a computer system. A username and, often, passwords are required in multi-user systems. A username is also required to access Siebel's online services.

User Primary Key
In a database table, one or more columns whose values produce a unique identifier for each row. Typically, these values have meaning for the user (such as first and last name). See also SYSTEM PRIMARY KEY.

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Validated Resource
A resource with a value that Siebel Quote Assistant prevents from becoming negative. See also UNVALIDATED RESOURCE.

Value
The quantity assigned to an element such as a variable, symbol, parameter, or label.

Variable
A symbol or name that stands for a value. Variables can represent numeric values, characters, character strings, and memory addresses.

Variable-Length Field
In a record, a field that can contain data of varying lengths.

Variable-Length Record
A record that has at least one variable-length field.

Variable Scoping
A term that refers to the level of accessibility of variables in Siebel VB scripts to other scripts. Levels of variable scoping include local, modular, global, and inter-application.

Variable Window
A window in the Siebel VB Debugger that displays the contents of the variables associated with a Siebel VB script when debugging.

Vendor Company
A company that sells its products through channel partner companies.

View
A display panel consisting of a particular arrangement of applets. The Siebel application window displays one view at any one time. You can change the currently active view by selecting a different view from the view bar or from a submenu of the Screens menu.

View Access Visibility
Also called responsibilities. The set of restrictions on a user's access to views in a Siebel application. The restrictions are based on the set of responsibilities associated with the user name. See also RESPONSIBILITY.

View Bar
The vertical column of view names that appears along the left edge of a window in a Siebel application. It contains the names of the views available in the currently selected screen. Click a name in the view bar to display that view.

View Designer
In Siebel Tools, a visual design tool used for assigning applets to single- or multiple-sector regions of a view.

View Object Type
In Siebel Tools, a visual design tool used for assigning applets to single- or multiple-sector regions of a view.

View Option
One of three presentations of the object hierarchy in the Object Explorer: The options are the Detail view option, Types view option, and Flat view option.

Visibility
Data access restrictions that define what views and data records each user can see. See also personal visibility, record access visibility, routing, sales team visibility, and view access visibility.

Visibility Event
A database change that causes a change in the visibility of related data.

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Web Page
1. A Siebel repository representation of a standard or configured HTML page that is part of an HTML application. 2. One page, or portion of a web site. 3. A document on the World Wide Web. Every web page is identified by a unique URL.

Web Search Specification
A specification that performs a dynamic lookup on a Web site based on a combination of special symbols and text. This is in contrast to a fixed URL, which navigates to a specific Web address. Also called a search specification URL.

Web Server
This server handles standard Web browser requests for HTML documents and other files. It also enforces security requirements by forcing users to log on and by allowing access to only those functions that users have permission to access.

Web Site
Usually a page or collection of pages, published on the World Wide Web by a business, organization, or person.

Web Template
A Siebel Repository representation of an HTML Template file. There is a one-to-one correspondence between the HTML Template file and an a Web Template Object. The Web Template contains information about the various Siebel Tags that appear in the HTML Template file, as well as references to other related HTML Template files.

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XYZ

XML
Acronym eXtensible Markup Language.

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